Introduction to Boolean modelling for biological systems

Computational Systems Biology for Digital Medicine
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Basic concepts and theory

- Computational systems biology methods have widely employed dynamic models to describe the biological functions from the dynamic system point of view.

- **Keywords:** system / dynamic

What is the advantage of adding a dynamic layer?

provide quantitative/qualitative descriptions of the network

predict the behavior of the network under different conditions, i.e., gene knockout, treatment with an external agent, etc.

Biological functions: Dynamic!

A biological function must be considered as a dynamic process

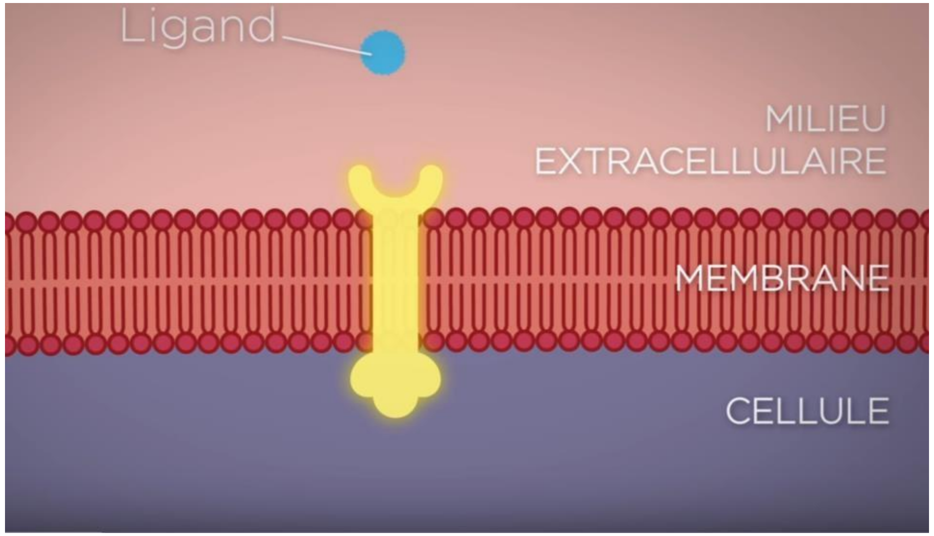
Within a cell, all the elements vary over time (energy production, respiration, formation of complexes, etc.)

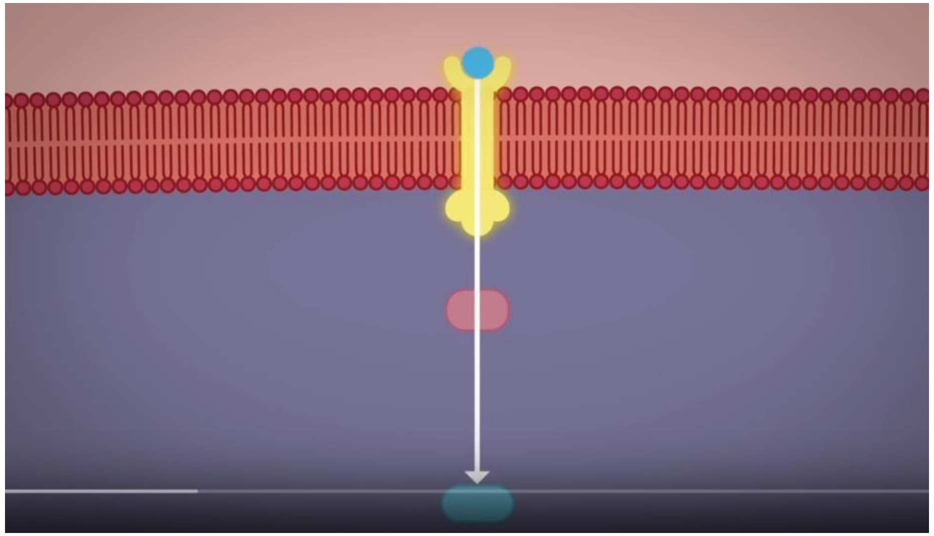
For our explorations: A cell frozen in time t

After or before: Definitely different function!

Many different layers

- Cellular functions involve multiple layers/ levels in the cell
- Functions cannot be defined independently of the context in which they operate

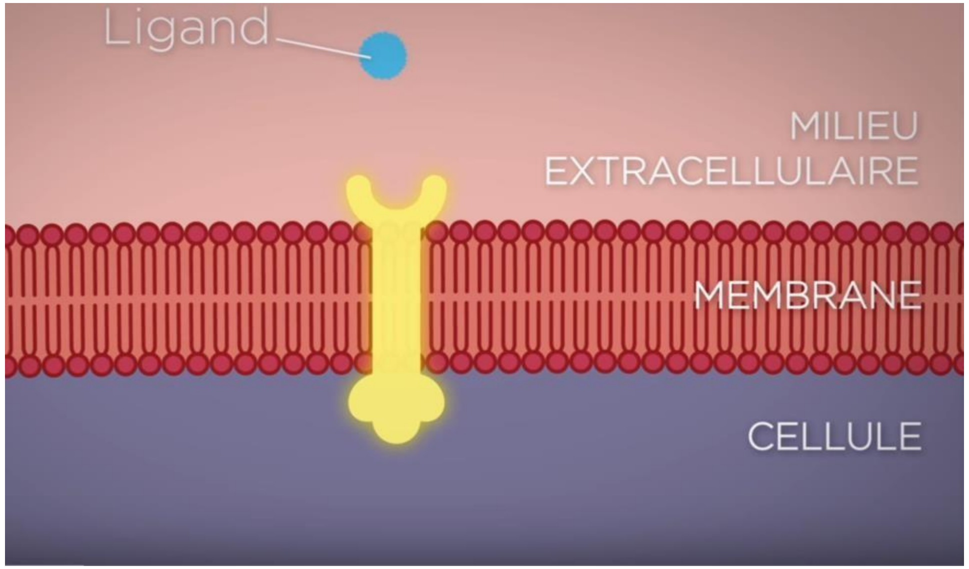


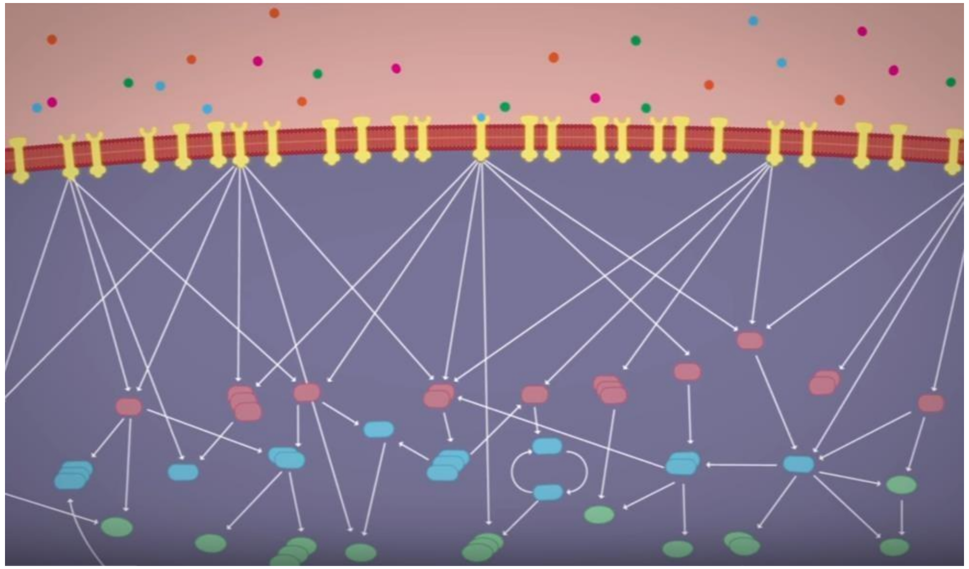


Several causes and consequences (effects)

- Several causes and several consequences coexist at the same time within the cell

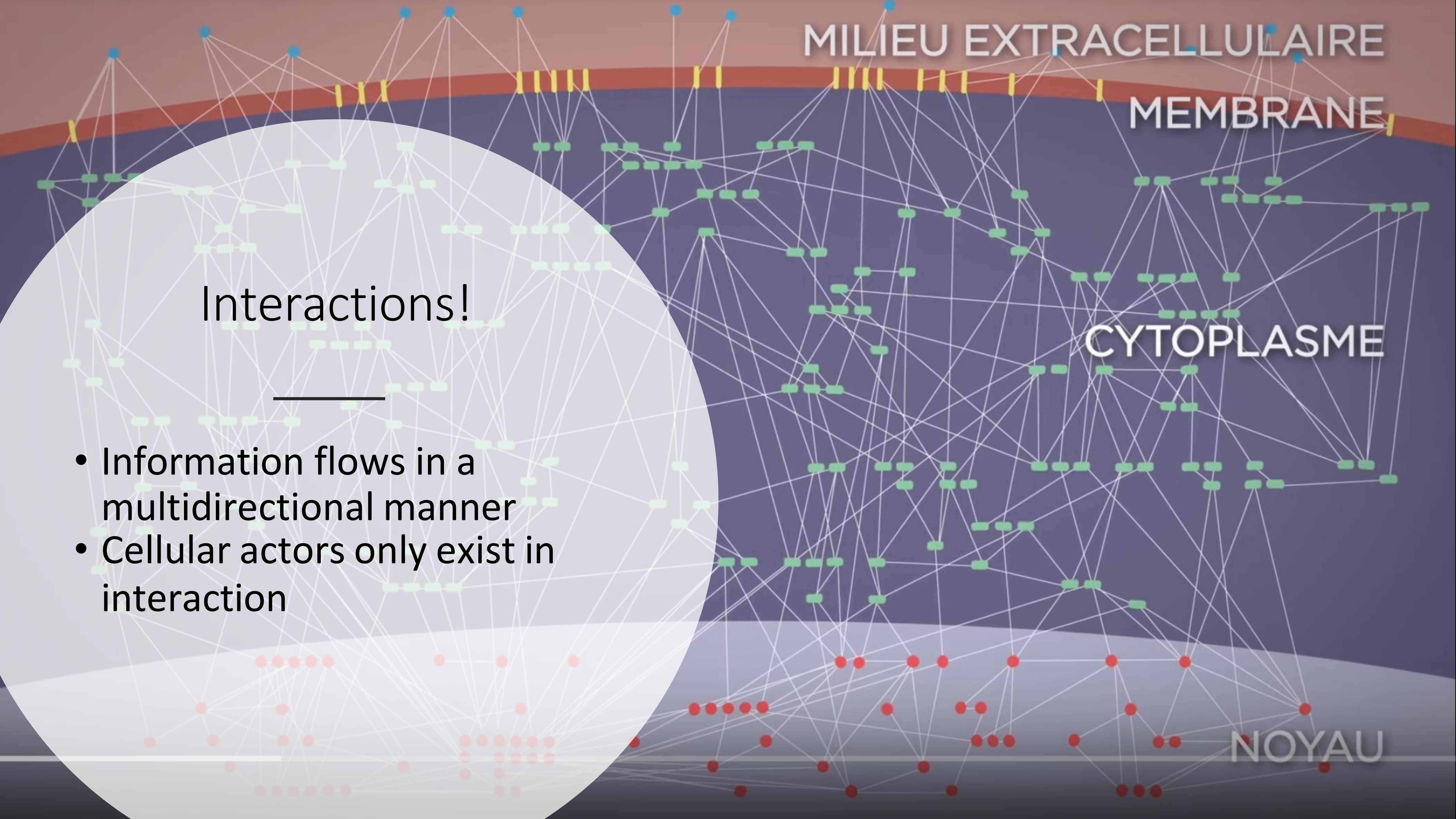
- In the context of cellular interactions, it is impossible to associate a single cause with each consequence and a single consequence with each cause





Interactions!

- Information flows in a multidirectional manner
- Cellular actors only exist in interaction



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Cellular context: complex!

COMPLEXITY OF CELLULAR CONTEXT:

ADOPT A DYNAMIC, SYSTEMS POINT OF VIEW TO ADDRESS IT

Boolean Algebra

- Boolean Algebra is a mathematical system for formulating logical statements with appropriate symbols, so that logical problems can be solved algebraically (in a manner similar to ordinary algebra).

- Boolean Algebra is introduced by an English mathematician, George Boole (1815-1864).

- The widespread use of Boolean Algebra is initiated by an American mathematician, Claude Shannon (1916-2001).

- He is credited with founding both digital computer and digital circuit design theory.

- George Boole

- Claude Shannon

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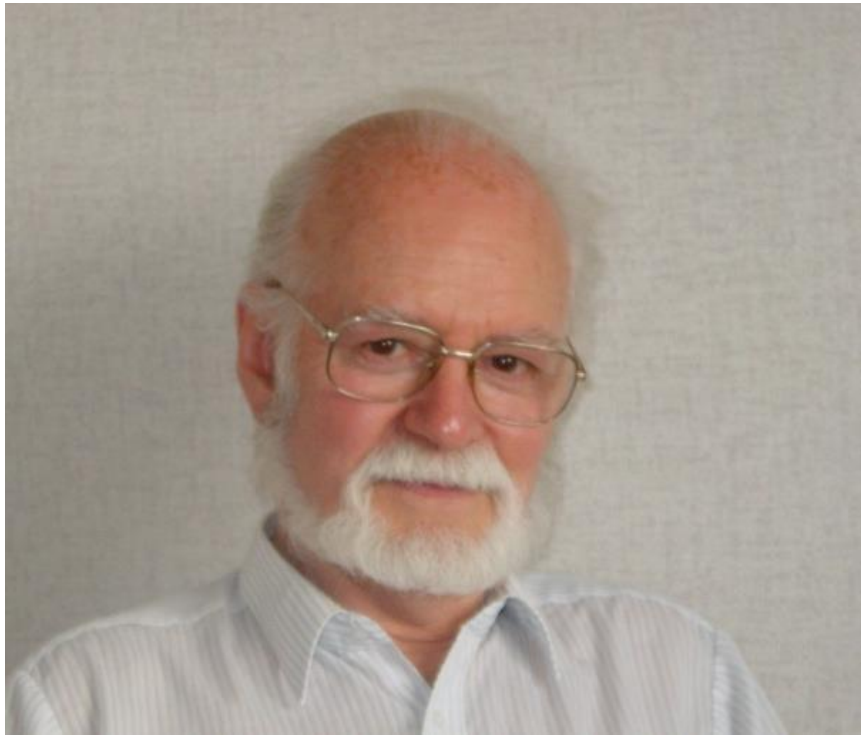
Boolean Algebra

- The variables in Boolean Algebra are the truth values of True (1) and False (0).
- The main operations of Boolean Algebra are :
  - The conjunction (**and**)
  - The disjunction (**or**)
  - The negation (**not**)

Boolean algebra for modelling regulatory networks

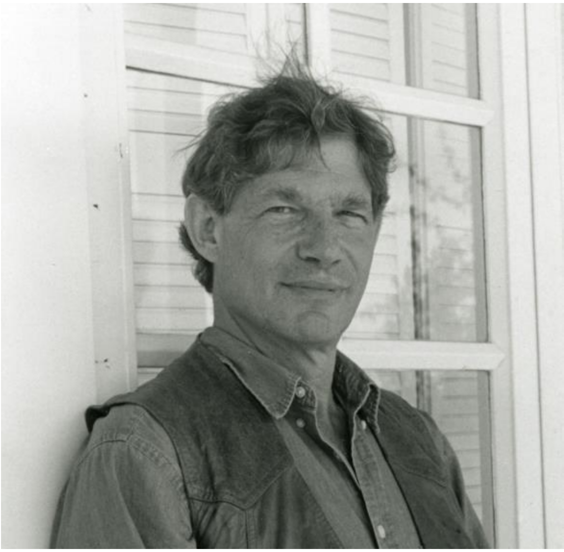
The experimental advances in the large-scale mapping of regulatory networks are fairly recent, but modeling efforts date back to the end of 1960s.

Pioneering work of Stuart Kauffman and Rene Thomas.



In the absence of experimental results, Stuart Kauffman considered an idealized representation of a typical gene network

He assumed that genes are equivalent, and their interactions form a directed graph in which each gene receives inputs from a fixed number K of randomly selected neighbors.



Focus on asymptotic behaviour

Two types of attractors: stable states or simple cycles

Deterministic behaviour (only one possible following state)
=> Synchronous strategy

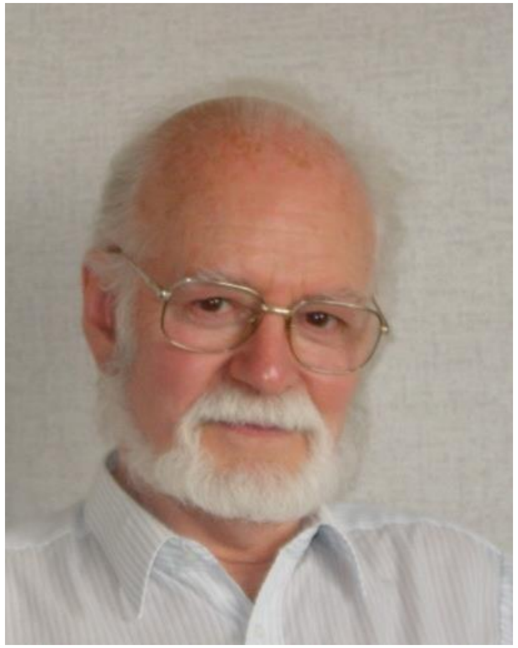
How does this work?

- The state of genes is described by binary (ON/OFF : 1/0) variables at a time point (t), and the dynamic behavior of each variable, that is, whether it will be ON (=1) or OFF (=0) at next moment (t+1), is governed by a Boolean function.

- A Boolean or logical function is written using the logical operators “and”, “or” and “not” to describe regulation.

Logical description of transcriptional regulation

René Thomas developed a detailed logical description of the mechanisms governing transcriptional regulation, including the effects of DNA domains such as promoters, initiators, terminators, and the concepts of genetic dominance and recessivity



Regulatory circuits: positive and negative

Non-deterministic behaviour (more than one possibility following state)
=> Asynchronous strategy

Feedback loops and multivariate variables

This formalism was later refined to include multilevel variables and used to study feedback loops, i.e. circular chains of interaction.

These loops can be classified into two categories based on the number of negative (inhibitory) interactions in the loop:

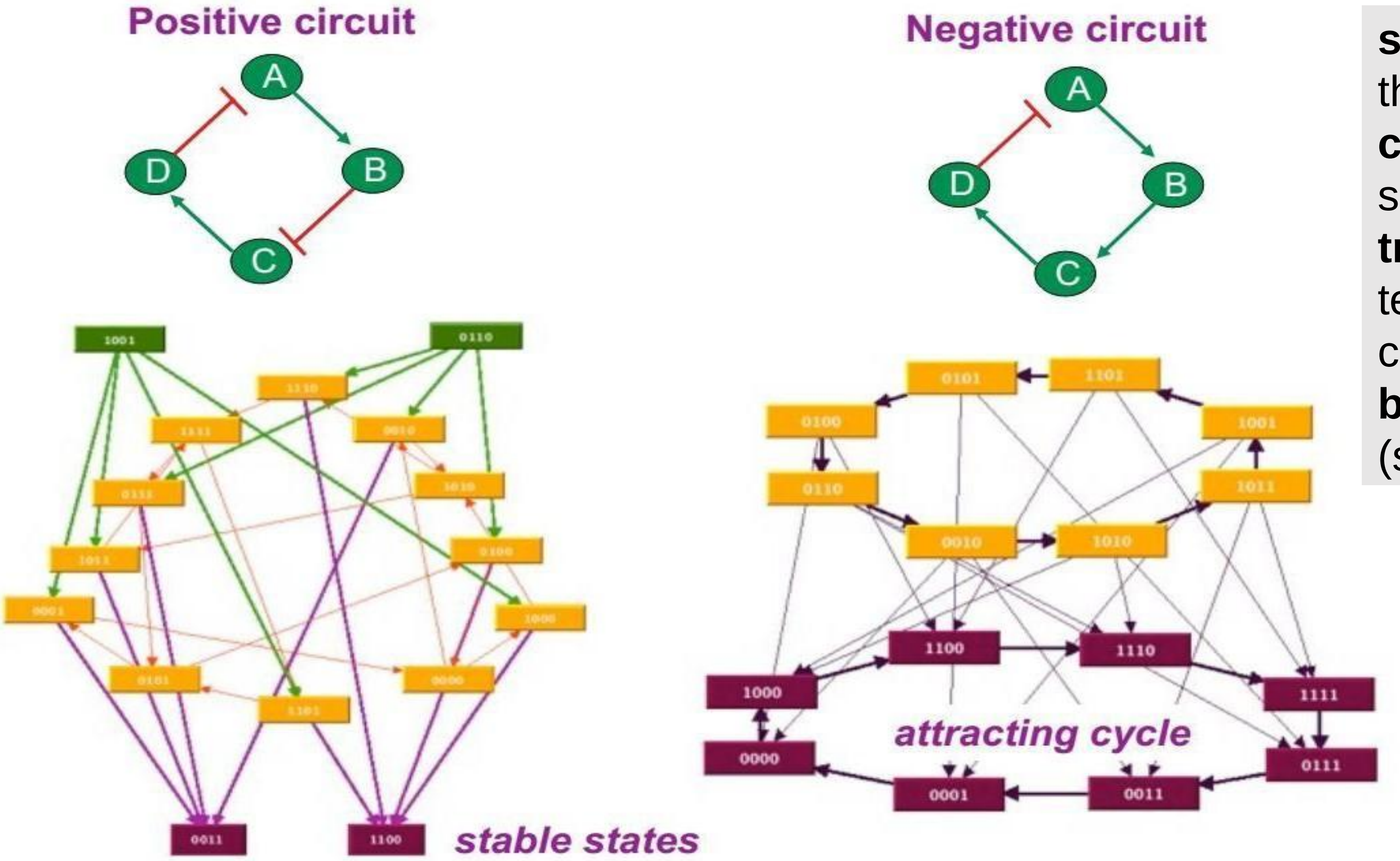
if this number is even, the loop is positive, and if the number of	negative interactions is odd, the loop is a negative feedback loop.

Associating feedback loops with dynamic behaviour and biological functions

Thomas found that a positive feedback loop is a necessary condition for the existence of multiple steady states, while a negative feedback loop with two or more elements is a necessary condition for stable limit cycles .

Biologically this means that cell differentiation is based on positive feedback loops, and homeostasis (stability to small perturbations) is based on negative feedback loops.

Regulatory circuits: dynamics in isolation



Remy *et al* (2003) *Bioinformatics* **10**: ii172-8

stable state (terminal nodes in the state transition graphs)
cyclic attractor (terminal strongly connected components)
transient cycle(s) (non-terminal strongly connected components)
basins of attraction (subsets of transient states)

When to use logical modelling?

- There is no quantitative information about the processes (rates of reaction, association/dissociation constants, quantities, etc.)

- The details of some interactions are unknown

- The biological question is qualitative (e.g. how does a cell chooses from a survival or an apoptotic phenotype?)

Discrete models: Boolean networks

Boolean networks have been widely used in modeling gene regulation

Switch-like behaviour of gene regulation resembles logic circuit behaviour

Conceptually easy framework: models easy to interpret

Boolean networks extend naturally to dynamic modeling

Boolean network modeling

- Boolean: either true or false (1 or 0)
- Binarization
  - reduces the noise in biological data
  - captures the dynamic behavior in complex systems
  - need a threshold value
  - leads to loss of information
- Biological entities are modeled as switch like dynamic elements
  - either on or off

Boolean networks

A Boolean network G(V, F) contains

- Nodes $V = \{x_1, \ldots, x_n\}$, $x_i = 0$ or $x_i = 1$
- Boolean functions

  $F = \{f_1, \ldots, f_n\}$
- Boolean function f_i is assigned to node xi

- Dynamic behaviour can be simulated

- State of a variable x_i at time t+1 is calculated by function f_i with input variables at time t

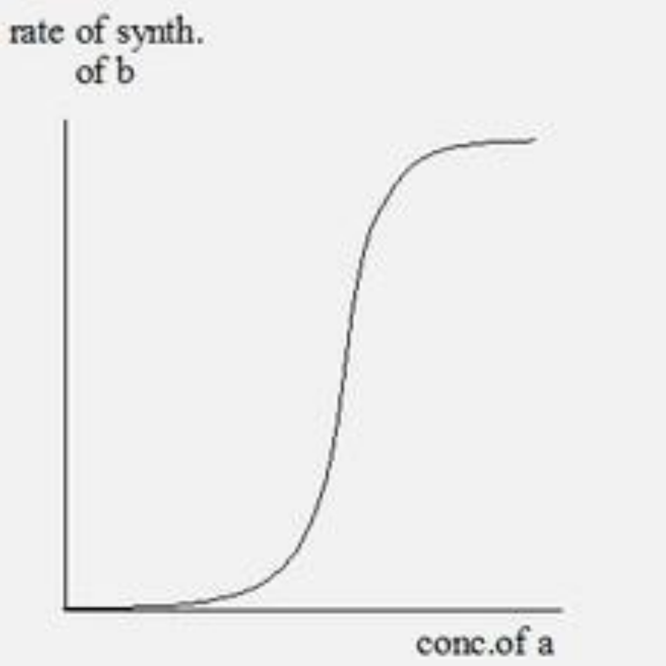
Logical Variable and Functions

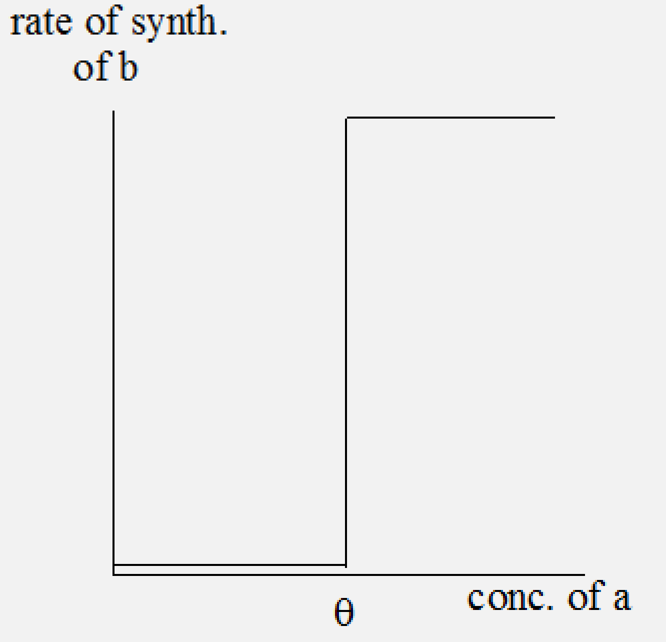
- Logical variables are associated with the elements of the system to describe the state of the system.
- They consist of the logical values. For example, a system whose state is appropriately described by the levels of substances a, b, and c, each of which can be absent, present at low level, or present at high level are represented by logical values 0, 1, and 2 respectively.

- If a product a acts to stimulate the production of b, it is a positive regulator. In this case, the rate of synthesis of b increases with increasing concentration of a, and makes a curve similar to that shown in figure A.
-

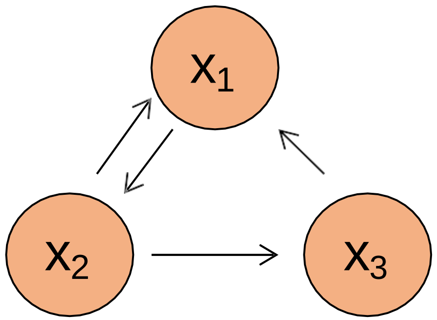
  There is little effect of a, until it reaches a threshold concentration theta, and at higher concentrations a plateau is reached which shows the maximal rate of synthesis of b.
- Such a nonlinear, bounded curve is called a sigmoid.

- It can be suggested that a is "absent" for a < theta and "present" for a > theta. The sigmoid curve can be approximated by the step function, as in figure B.





Example Graph (G) with 3 genes



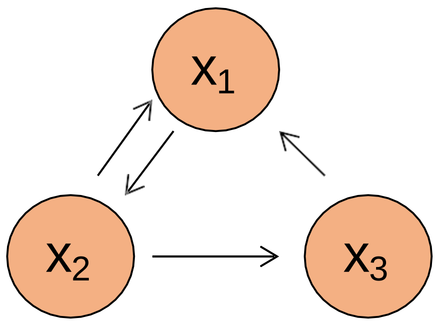
Wiring diagram

G(V,F),
$V = \{x_1, x_2, x_3\}$
$F = \{f_1 = x_2 \& x_3,\ f_2 = x_1,\ f_3 = x_2\}$

Input (t-1)			Output(t)		
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

Truth table

Example Graph (G) with 3 genes



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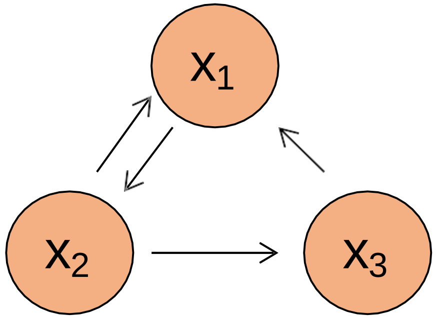

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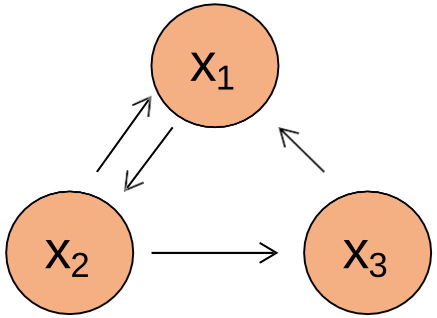

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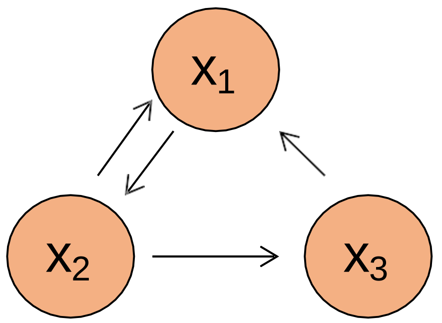

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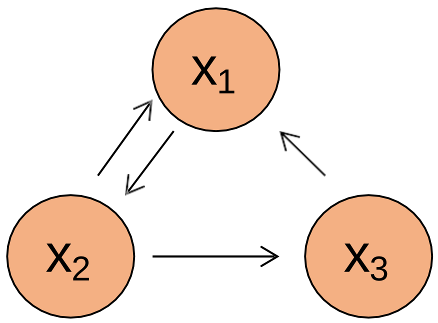

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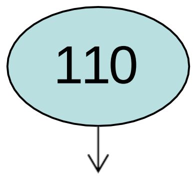

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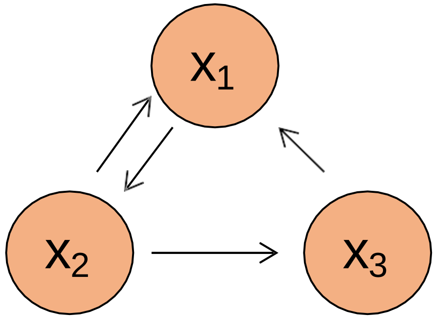

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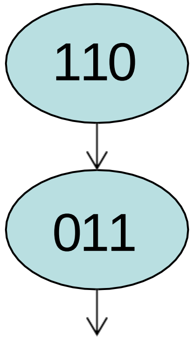

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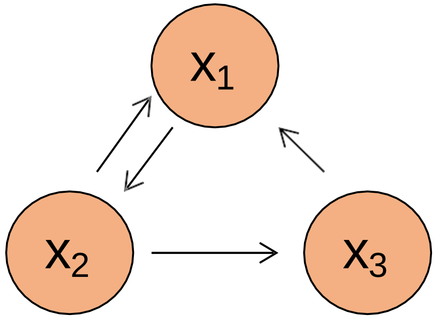

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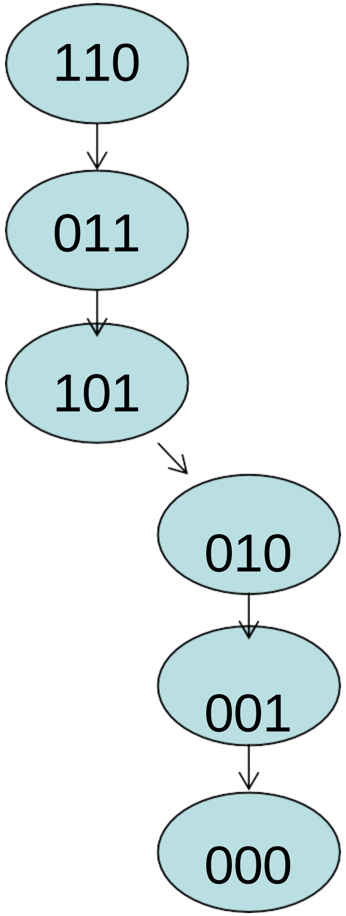

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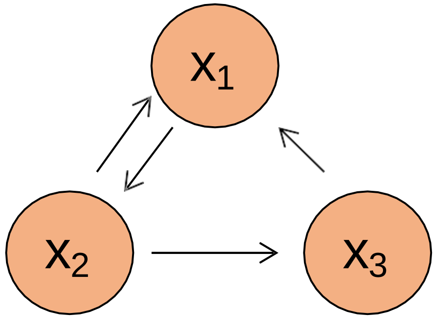

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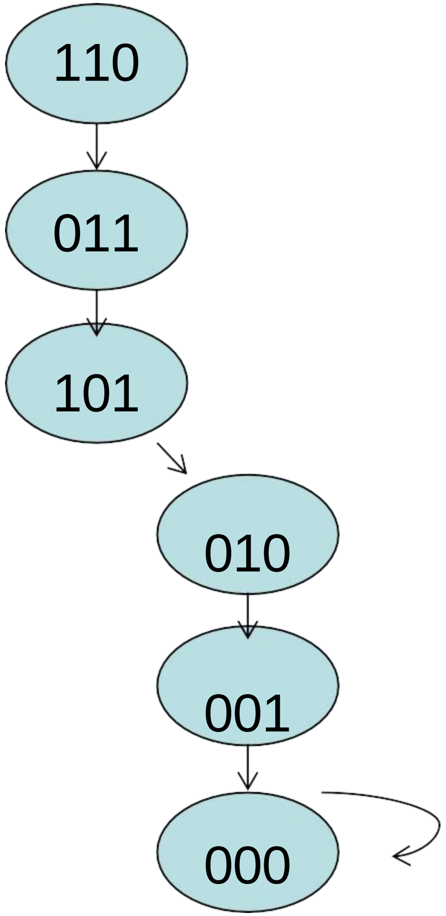

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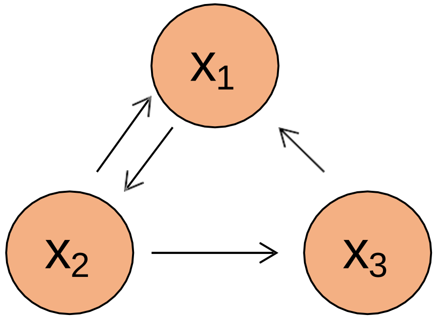

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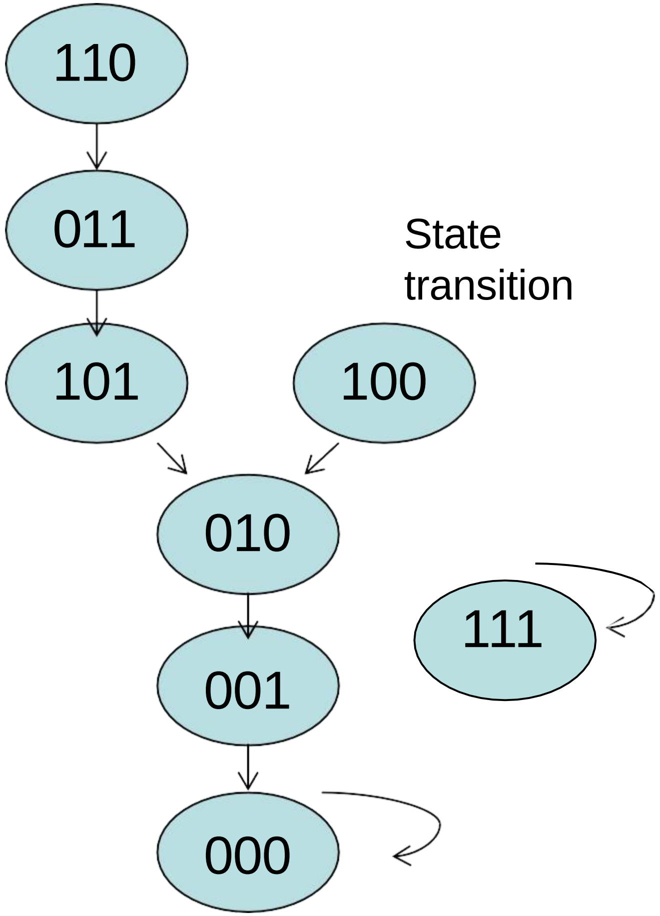


Logic -based models

- Form of mathematical model governed by logic operators (AND, OR, NOT).
- Parameter free.
- Suitable for modeling gene regulatory networks.
- *In silico* simulations, qualitative predictions.
- Each node in a logic model has a corresponding logic function that controls its regulation each time the model is updated.
- Two updating schemes: synchronous and asynchronous.

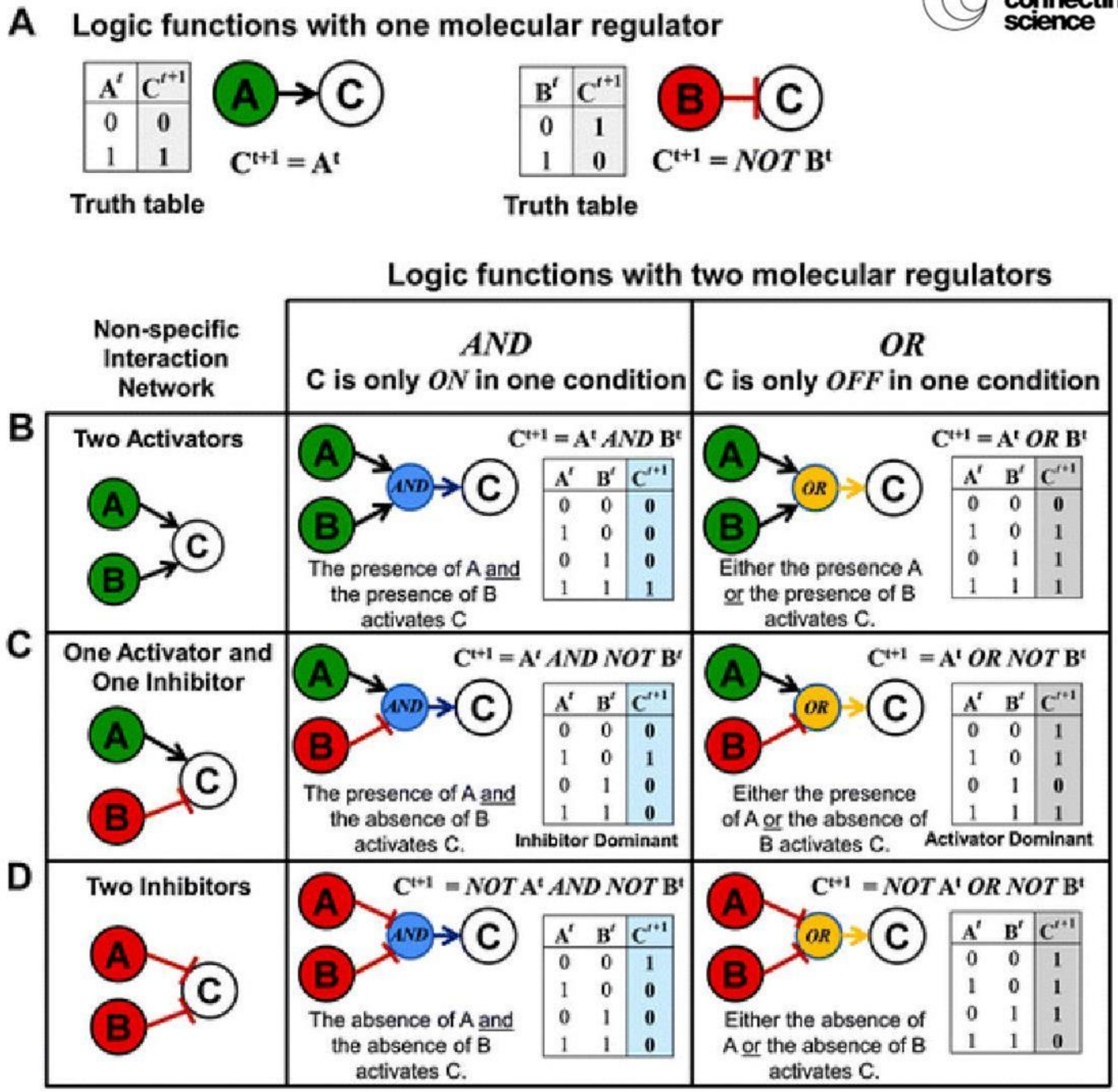
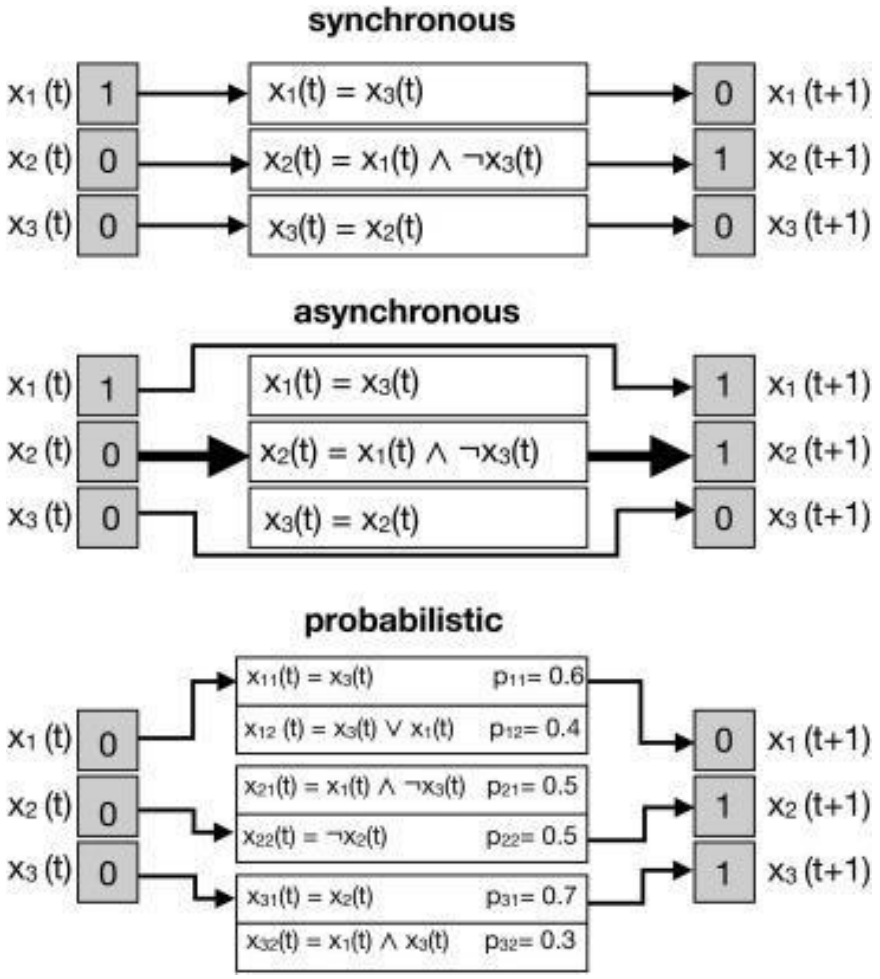


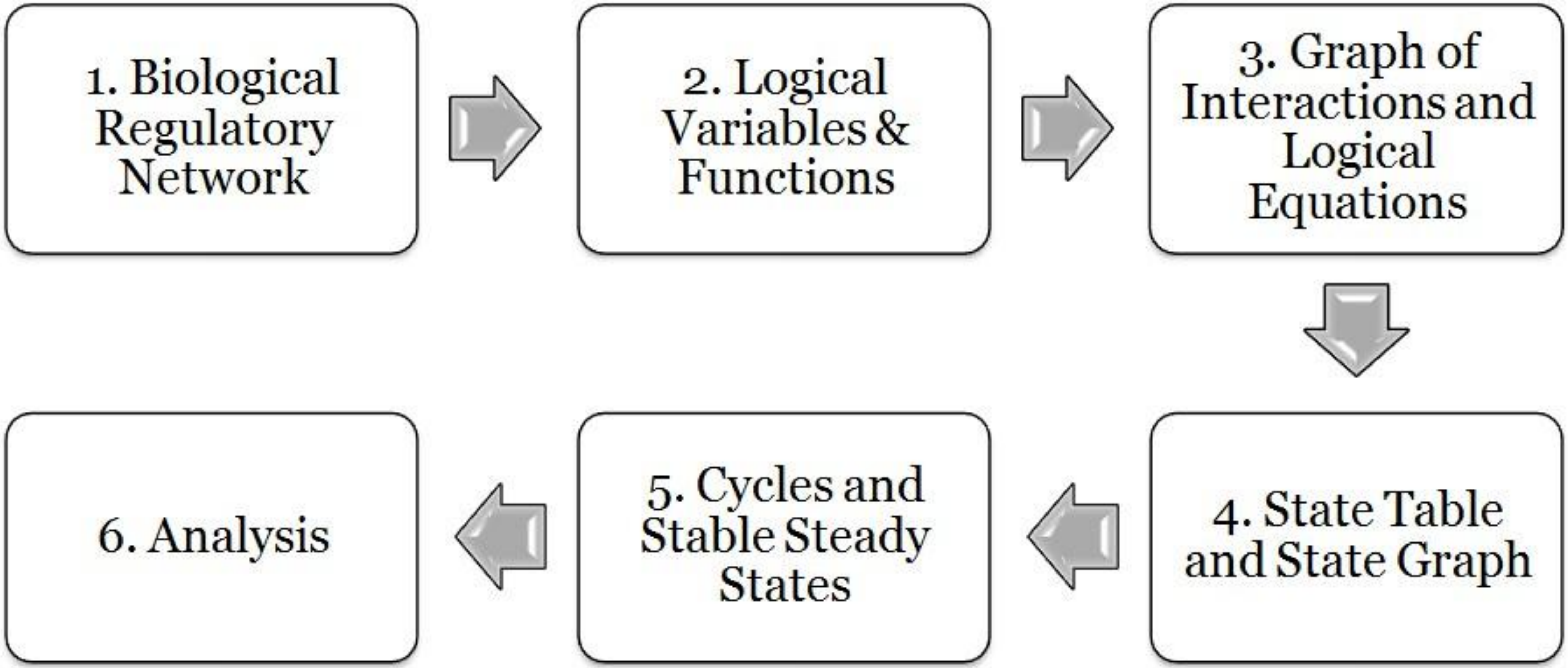
Figure from Wynn et al, 2010

Updating schemes

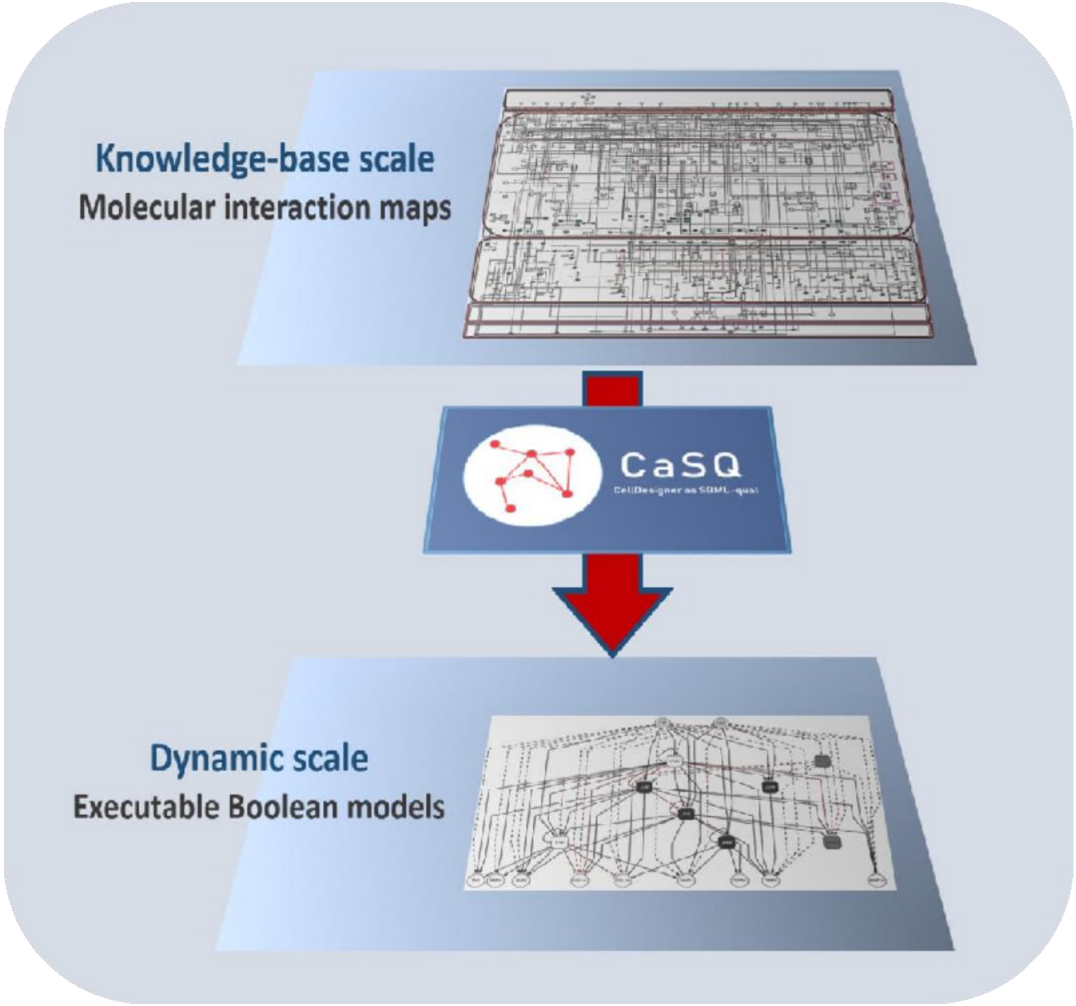


In synchronous models all Boolean functions are applied at the same time while in asynchronous models only one randomly chosen function (fat arrow) is updated per step.

Probabilistic models can have multiple functions with a predefined probability. One function per variable x_i is randomly chosen in each time step and then synchronous updating is performed


1. Biological Regulatory Network
2. Logical Variables & Functions
3. Graph of Interactions and Logical Equations
4. State Table and State Graph
5. Cycles and Stable Steady States
6. Analysis

From Static Representations To Dynamic Models



CellDesigner™

<u>Input</u>: CellDesigner XML file

CaSQ [1]
CellDesigner as SBML-qual

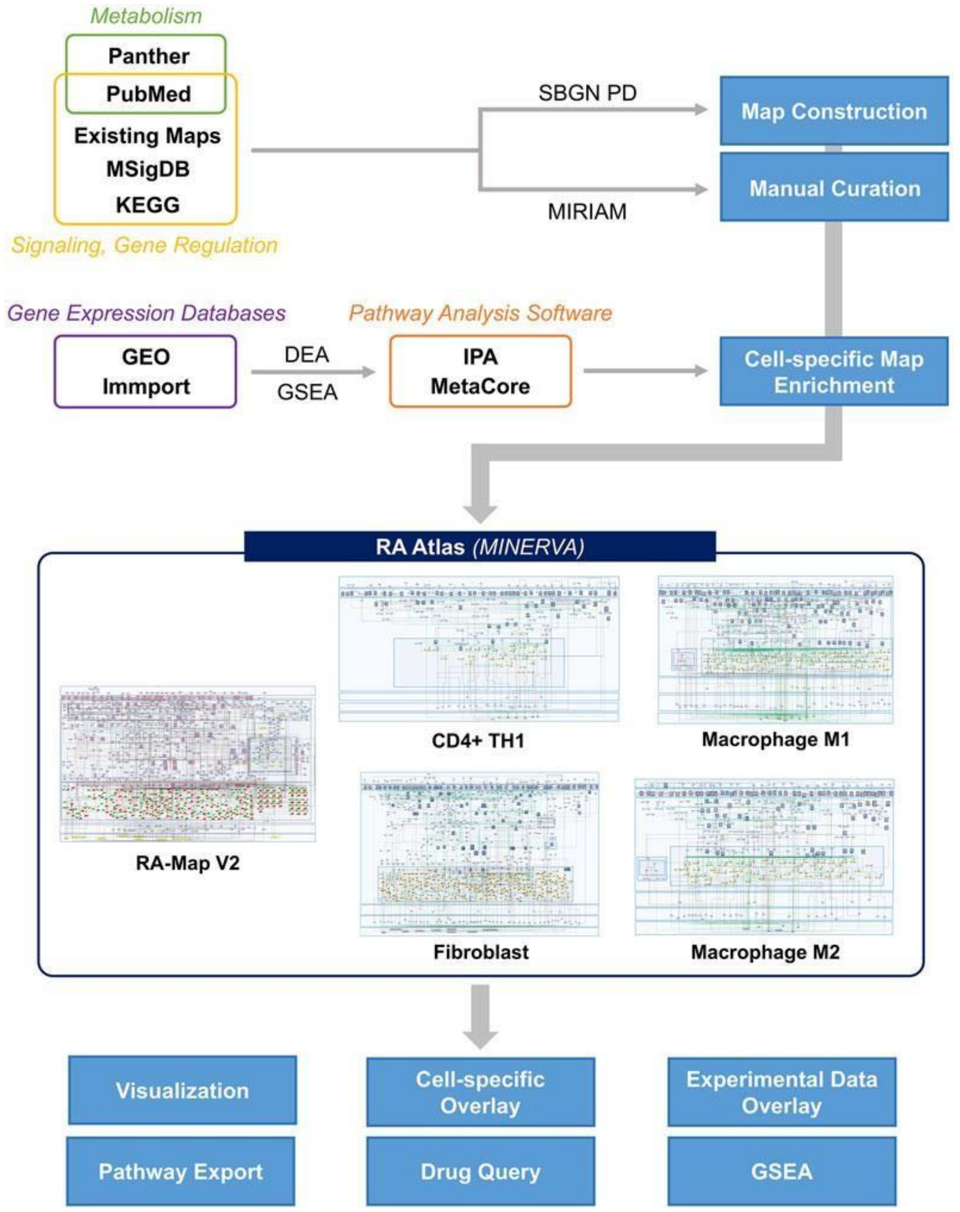
Python package

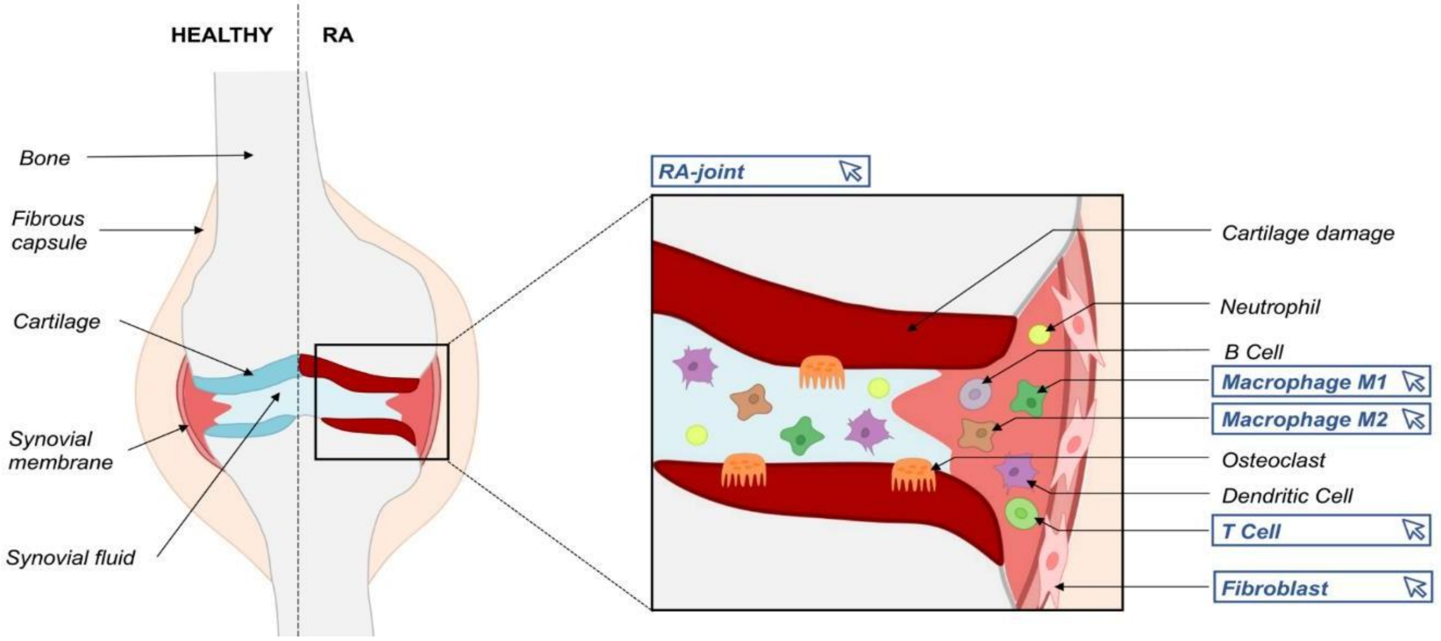
SBML™

<u>Output</u>: SBML-Qual file

[1] Aghamiri S. S., Singh V., *et al*. **Automated inference of Boolean models from molecular interaction maps using CaSQ**. *Bioinformatics* (2020).

Concrete application

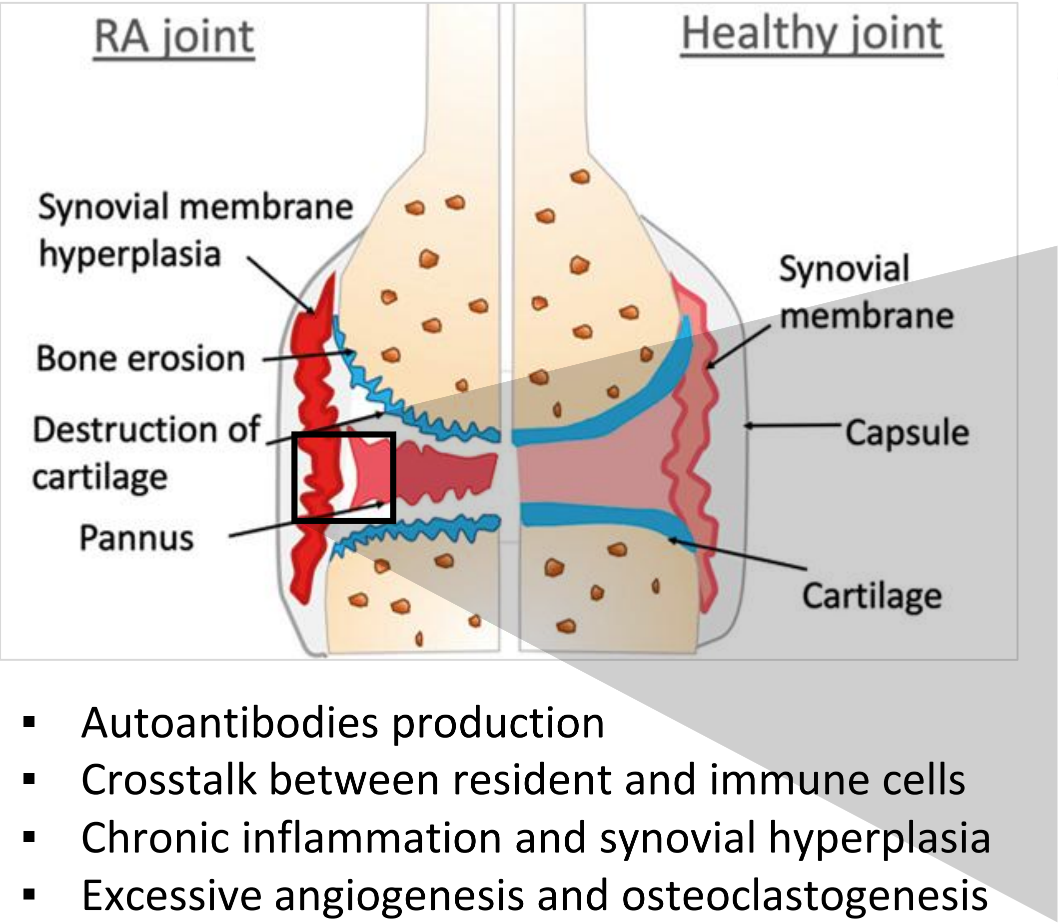




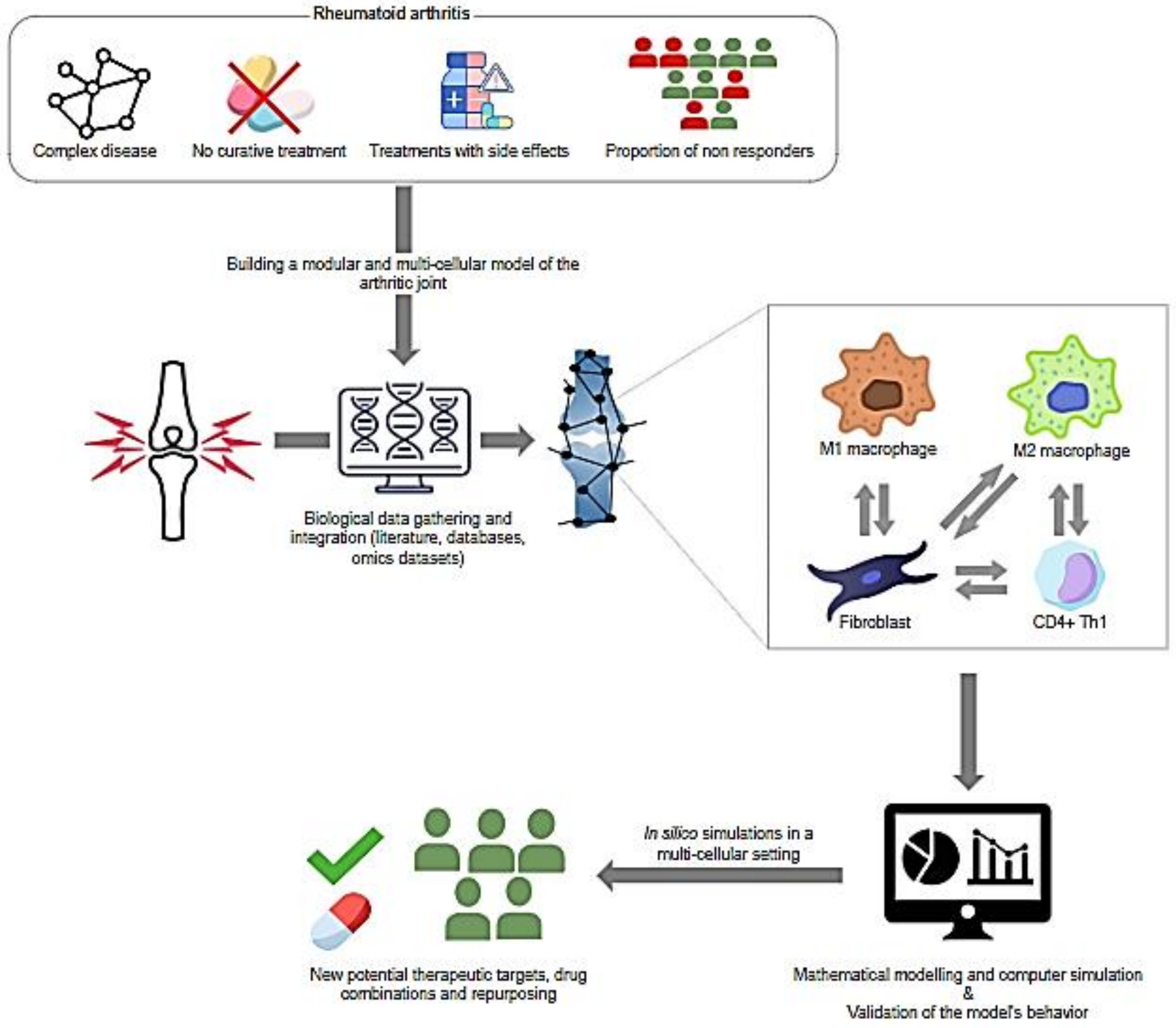
Maps	Species	Reactions	Phenotypes
Fibroblast	853	509	9
M1 macrophage	640	448	8
M2 macrophage	520	342	7
CD4+ Th1	321	179	7

(**Zerrouk** et al., 2022)

Rheumatoid arthritis

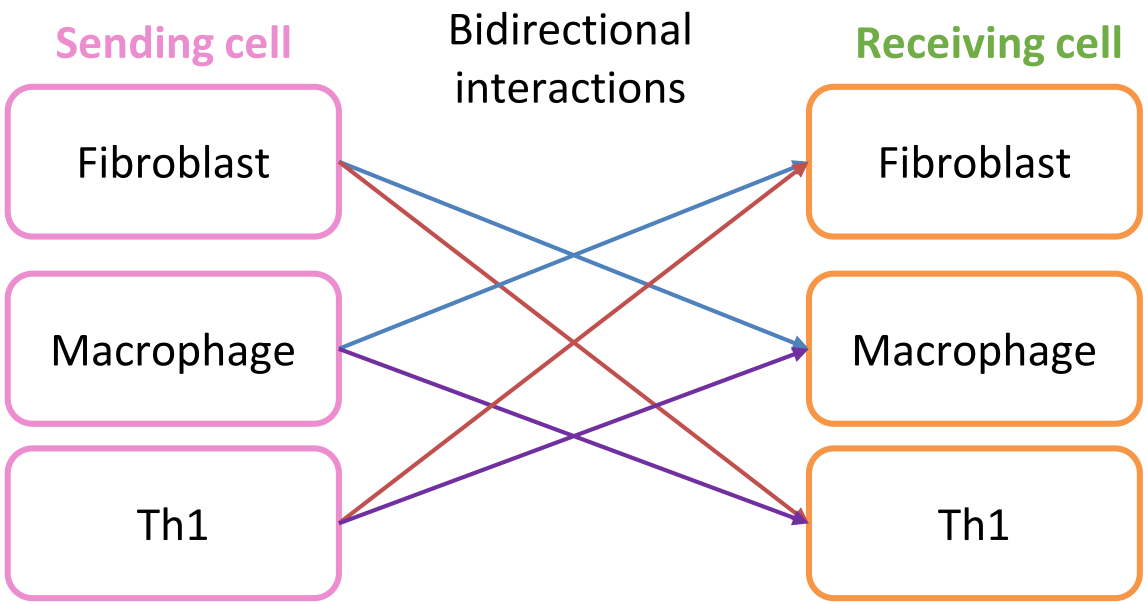


- Autoantibodies production
- Crosstalk between resident and immune cells
- Chronic inflammation and synovial hyperplasia
- Excessive angiogenesis and osteoclastogenesis
- Cartilage destruction and bone erosion



(Walsh and Choi, 2014)
(Isabel Lopez-Oliva et al., 2019)

Datasets selection & filtering criterias for the identification of cell-cell communication



- Three different scRNA datasets (only 1 dataset containing the three synovial cell types)
- Cross validation via the combination of datasets coming from different experimental studies
- ComBat method to correct the batch effect (Zhanh et al. 2020, NAR)

Filtering criterias

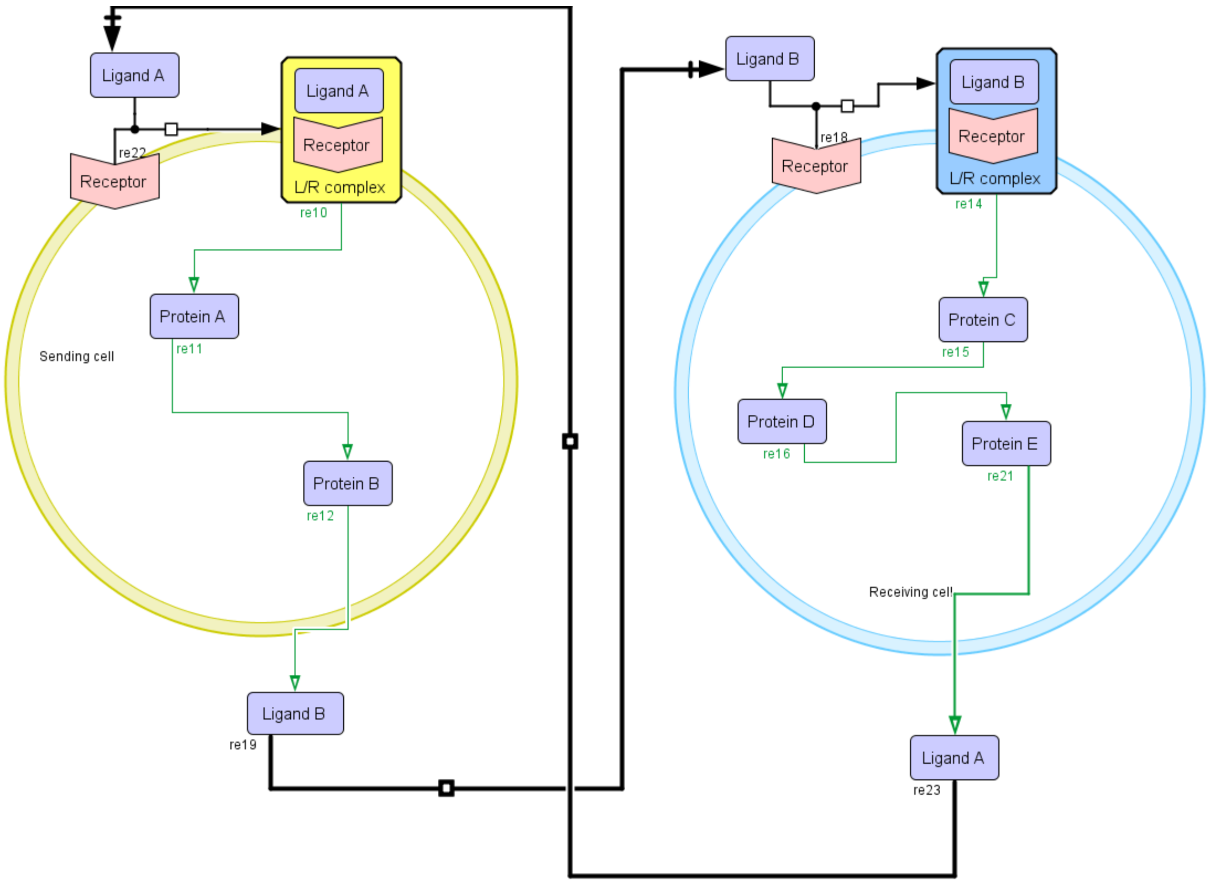
- Statistically significant adjusted p-values
- Both in ICELLNET and DiSiR
- Associated with at least two different pairs of datasets, or two different approaches
- Associated with at least two different sources of information

Omics datasets used in this project

Dataset	Technology	Database	Cell type(s)	Outcomes
GSE97779	Microarray	GEO	-RA synovial macrophages -Healthy macrophages	-DEGs between RA and healthy macrophages with preprocessCore, BioMart, Limma FDR < 0.05 & abs(FC) > 1.5
GSE164498	Single cell RNA seq	GEO	-Healthy M1 macrophages -Healthy M2 macrophages	-M1 and M2 marker genes with Bioturing and Venice method Classifier's FDR < 0.05
GSE109449	Single cell RNA seq	GEO	-RA synovial fibroblasts -OA synovial fibroblasts	-DEGs between RA and OA fibroblasts with Bioturing and Venice method FDR < 0.05 & abs(FC) > 1.5
SDY998	Single cell RNA seq	Immport	-RA synovial fibroblasts -RA synovial monocytes -RA synovial T cells -RA synovial B cells	-Synovial CD4+ Th1 cluster with Find_neighbors and Find_clusters functions in Seurat + ABIS shiny app -DEGs between RA Th1 cluster and RA naive CD4+ T cells using Seurat and Find_marker_genes function FDR < 0.05 & abs(FC) > 1.5 -Gene sets enriched in the synovial Th1 cluster vs naïve CD4+ T cells using IPA gene sets and GSEA software FDR < 0.05
GSE32901	Microarray	GEO	-Healthy PBMCs (including naive CD4+ T cells and Th1 cells)	-Gene sets enriched in the Th1 cells from PBMCs samples using IPA gene sets and GSEA software FDR < 0.1
GSE107011	RNA seq	GEO	-Healthy PBMCs (including naive CD4+ T cells and Th1 cells)	-Gene sets enriched in the Th1 cells from PBMCs samples using IPA gene sets and GSEA software FDR < 0.1
GSE135390	RNA seq	GEO	-Healthy PBMCs (including naive CD4+ T cells and Th1 cells)	-Gene sets enriched in the Th1 cells from PBMCs samples using IPA gene sets and GSEA software FDR < 0.1
E_MTAB_8322	Single cell RNA seq	ArrayExpress	-RA synovial macrophages	-Normalized gene expression in the RA synovial macrophages using Seurat's NormalizeData function -The five treatment naïve RA patients only

Construction of the RA multicellular map

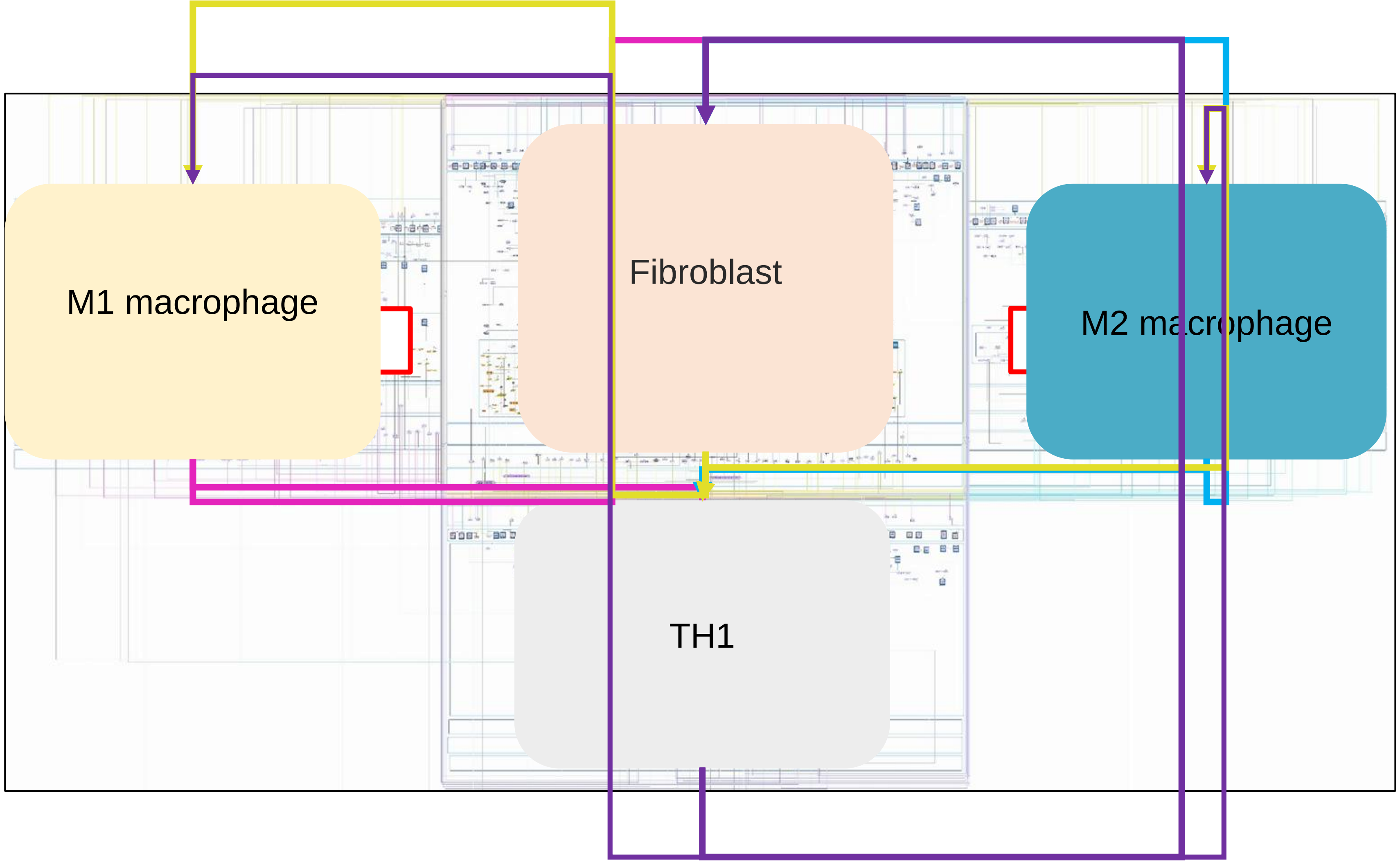
- **The black arrow**: transport of the secreted ligand
- Yellow cell : Sending and Receiving cell
- Blue cell : Sending and Receiving cell



Schematic representation of the intercellular interactions

The RA multicellular map

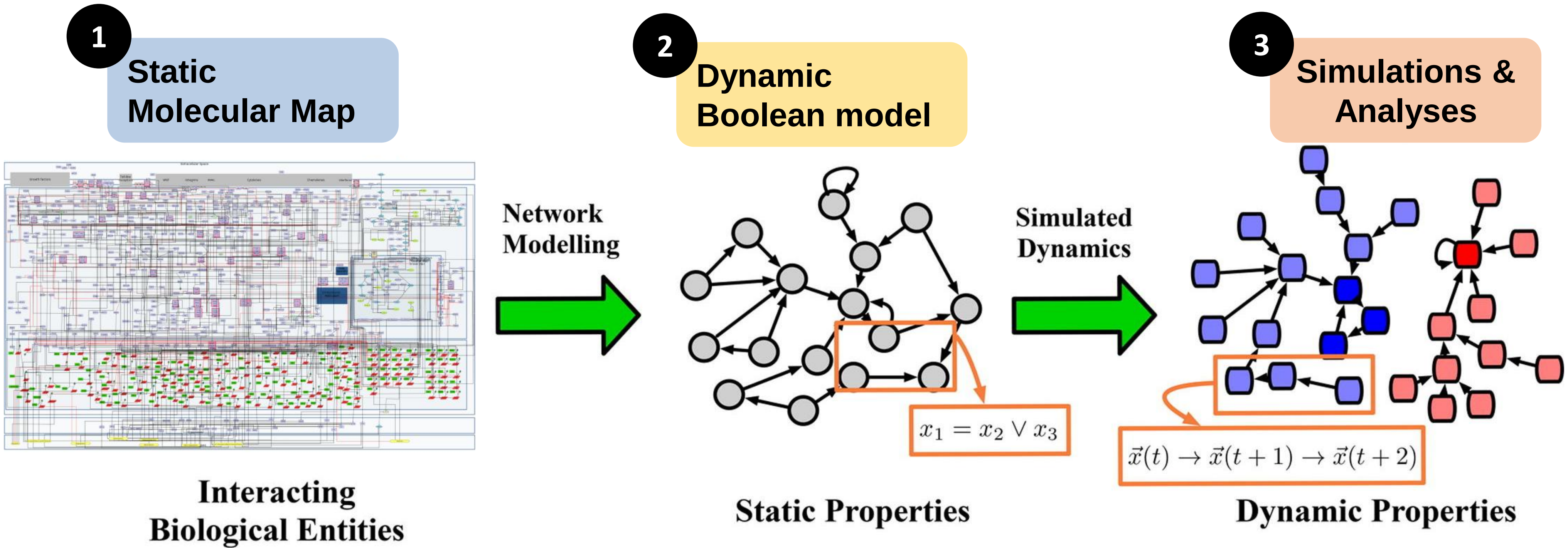
- **2232** components that interact with one another via **1461** reactions including **118** intercellular interactions



Color coding:

- M1 macrophage
- M2 macrophage
- Fibroblast
- Th1

Static and dynamic representations of causal networks

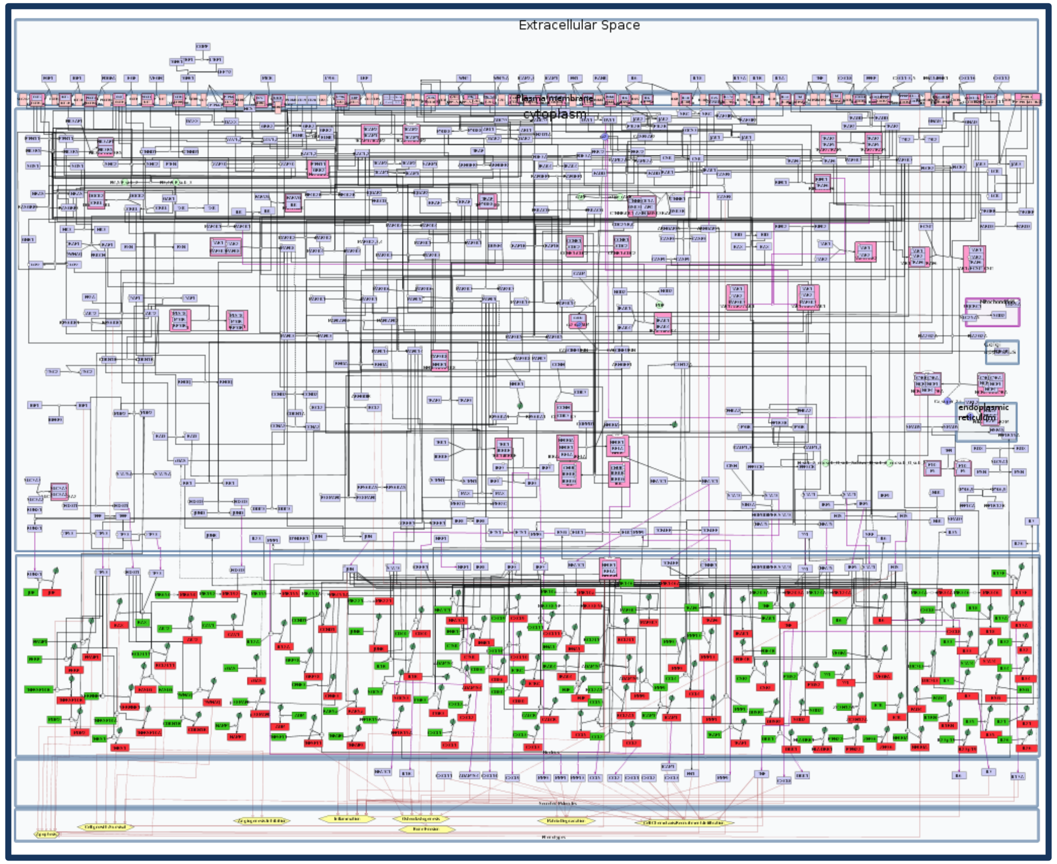


- Node: 0 or 1 → biological entity absent/inactive or present/active
- Edges: activators or inhibitors
- Activation of one or multiple nodes → Model behavior

Adapted from Schwab et al., 2020

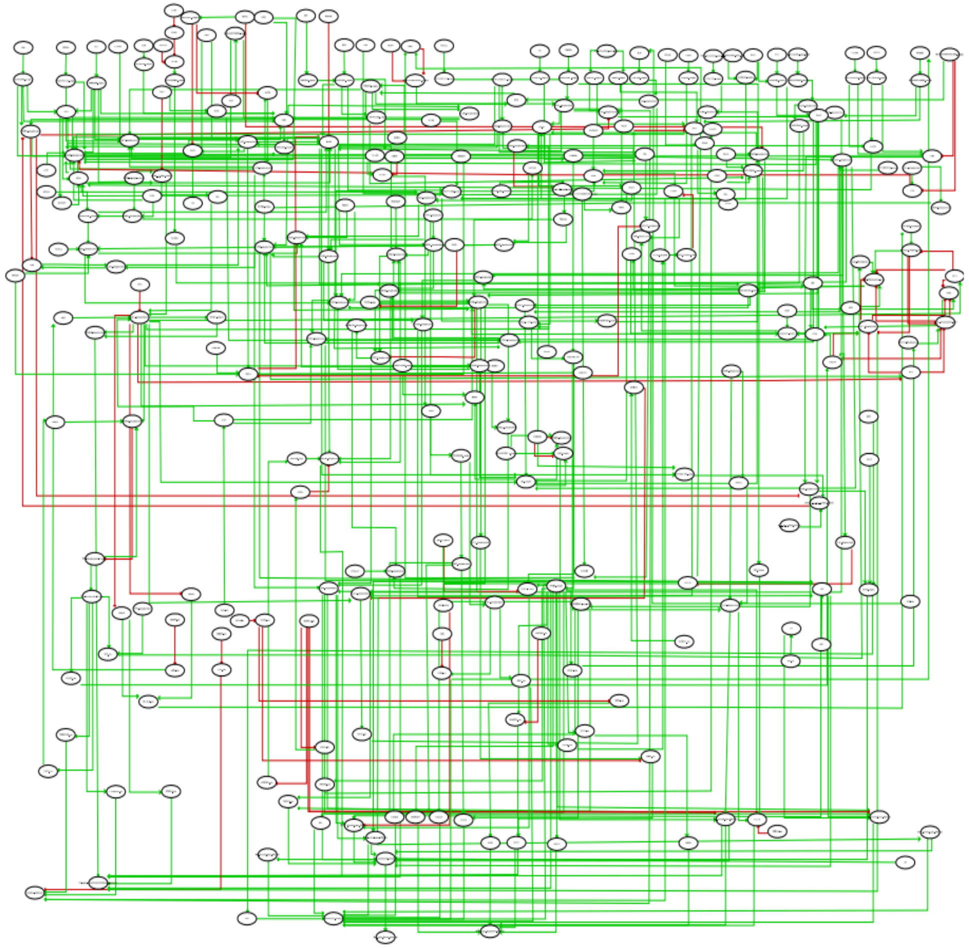
Application

PD Disease Map



CaSQ

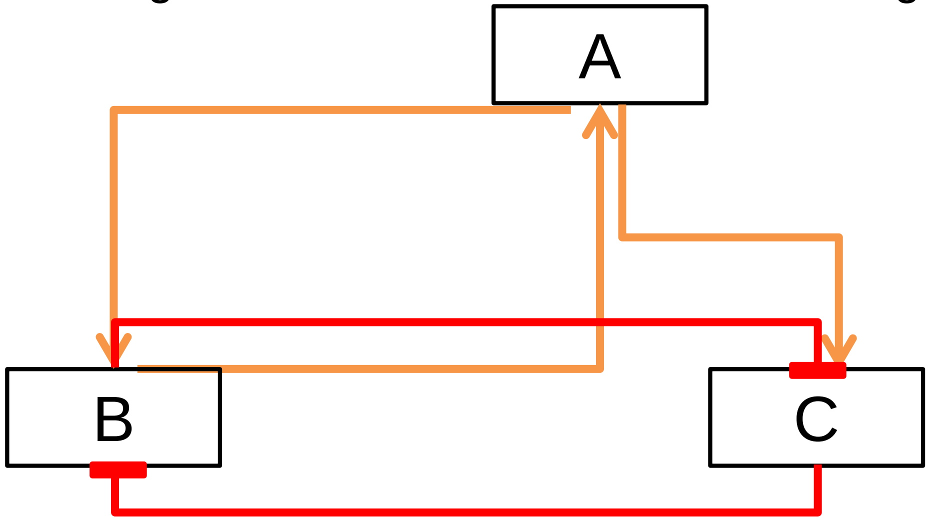
Boolean model



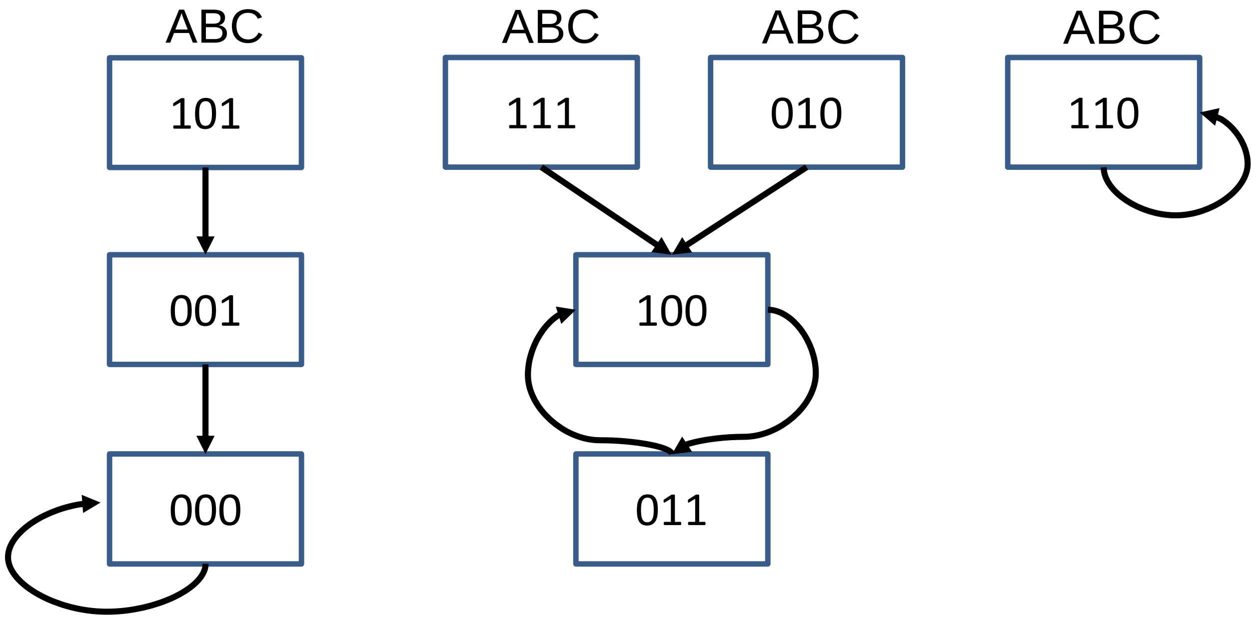
V. Singh, M. Ostaszewski, G.D Kalliolias, G. Chiocchia, R. Olaso, E. Petit-Teixeira, T. Helikar, A. Niarakis Computational Systems Biology Approach for the Study of Rheumatoid Arthritis: From a Molecular Map to a Dynamical Model. Genomics and Computational Biology Vol 4 No 1 (2018)

State graphs and attractors

- State graph : the transition of one variable from one point in time to the next
- Synchronous and asynchronous updating schemes
- **In synchronous models all Boolean functions are applied at the same time while in asynchronous models only one randomly chosen function is updated per step.**
- Long-term behavior = attractors = biological phenotypes



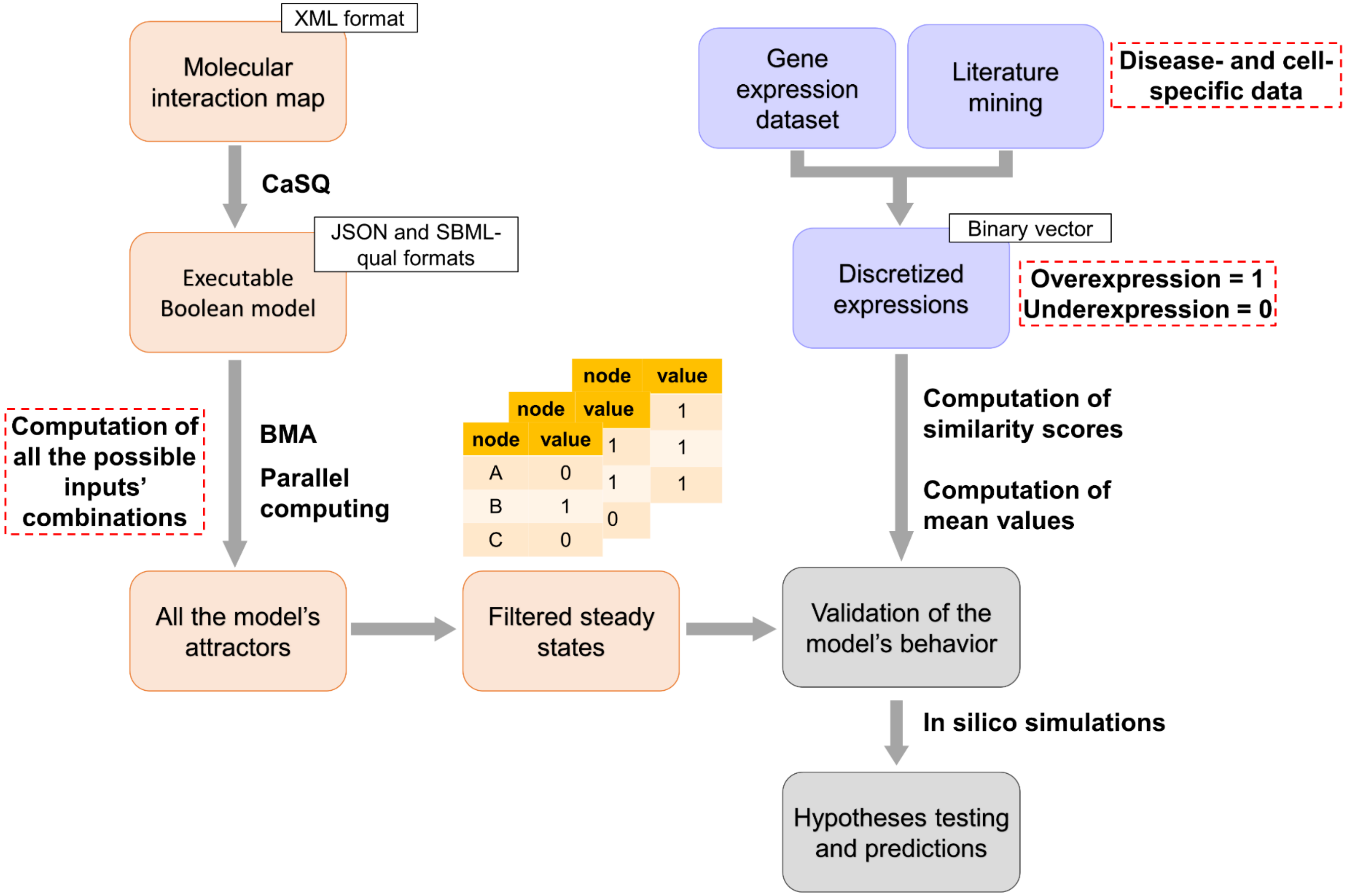
A = B
B = A AND NOT C
C = A AND NOT B



State transition graph and attractors of the toy model

(Schwab et al., 2020)

The proposed computational framework



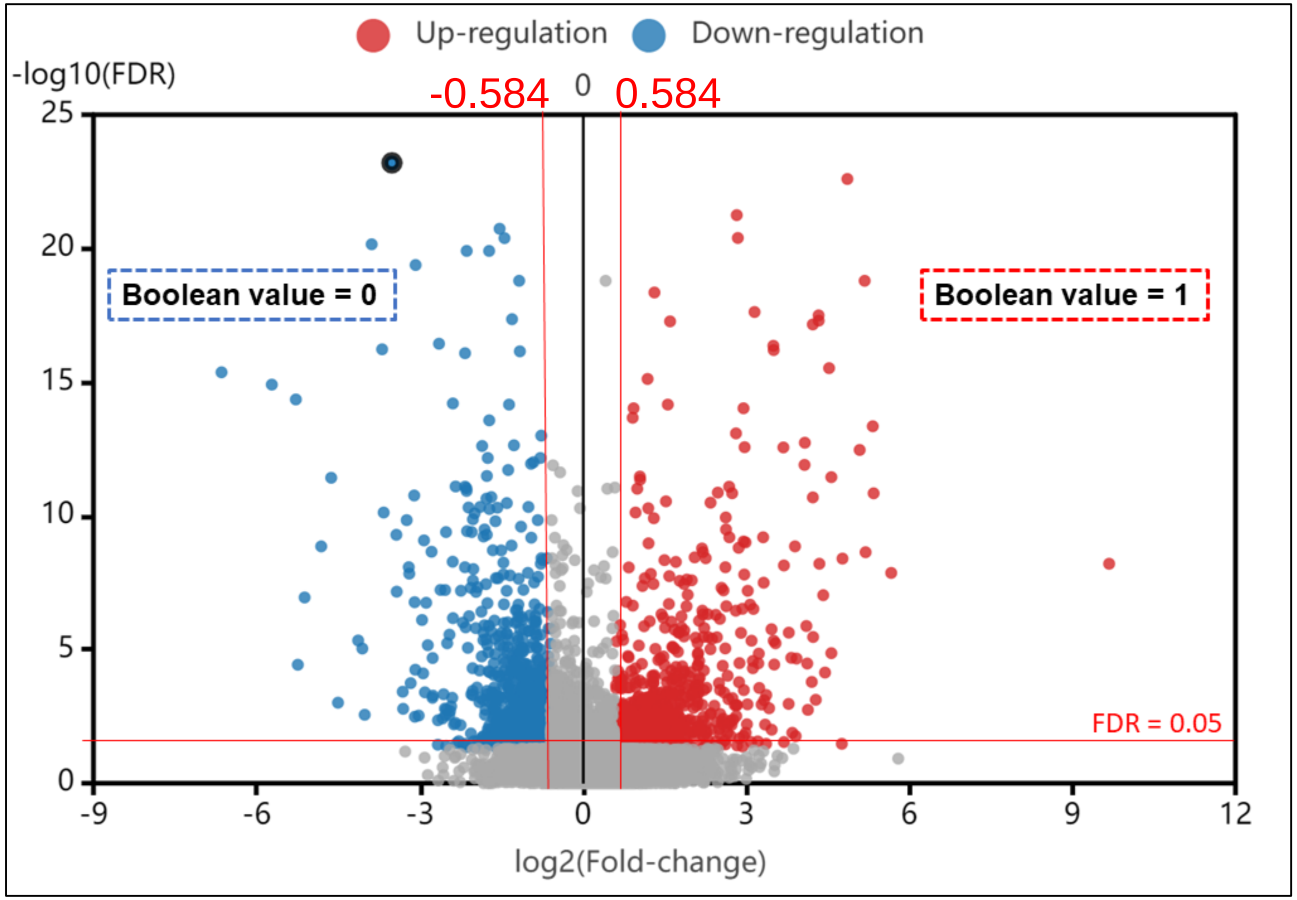
(**Zerrouk** et al., npj systems biology & applications, 2024)

Validation of the model's behavior

- Similarity score S = $(N00 + N11)/(N00 + N11 + N10 + N01)$

N00 and N11 = number of nodes **with the same state** in both the steady state and the discretized vector of experimentally observed expressions

N01 and N10 = number of nodes with **different states** in the steady state and the discretized vector of experimentally observed expressions

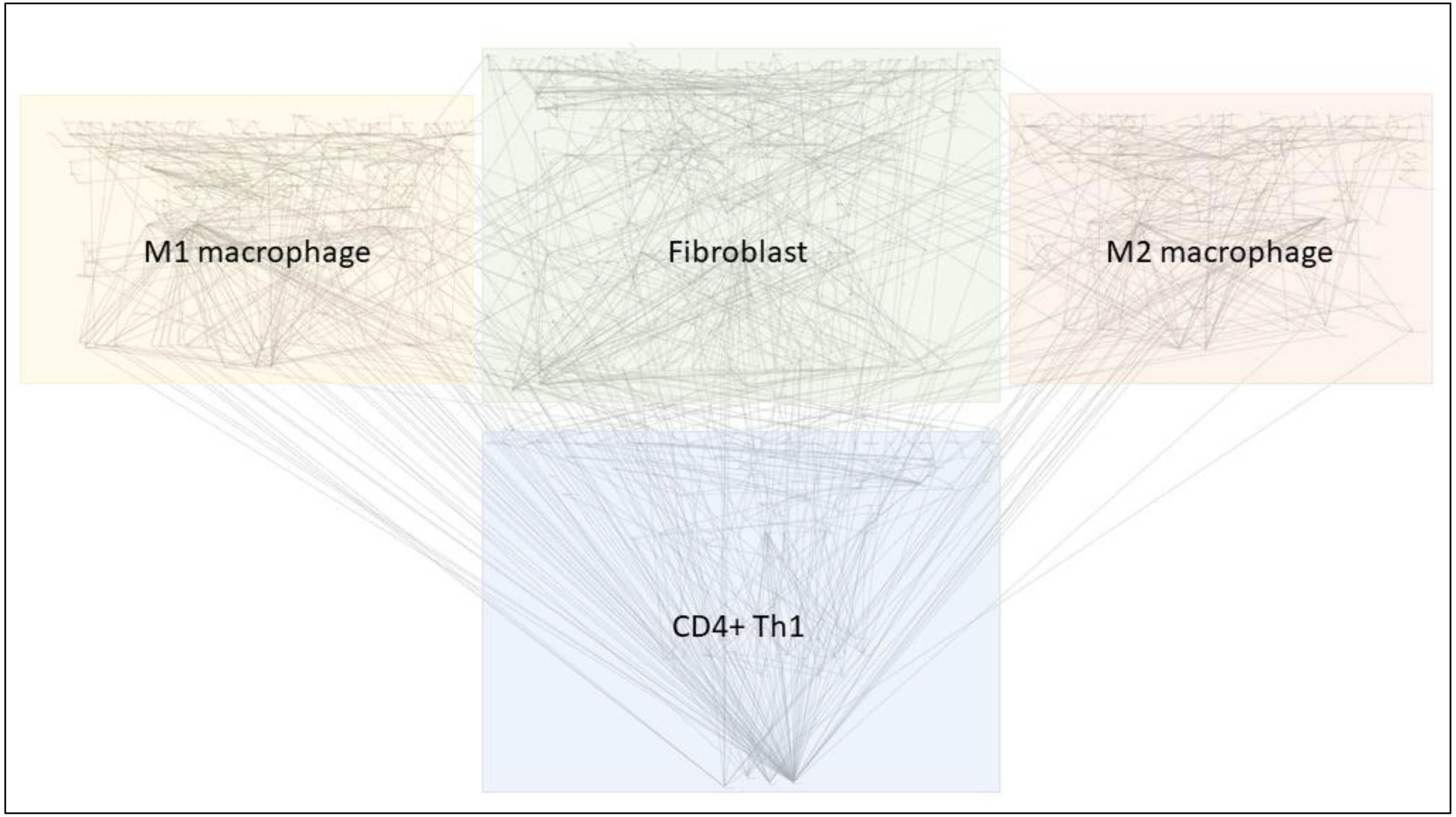


Gene expression discretization on a volcano plot showing the DEGs between RA and osteoarthritis synovial fibroblasts. DEGs were filtered using an FDR equal to 0.05 and a logFC equal to 0.584.

(**Zerrouk** et al., npj systems biology & applications, 2024)

From the RA multicellular map to a multicellular executable model

1104 nodes, including 240 inputs, and 1845 interactions

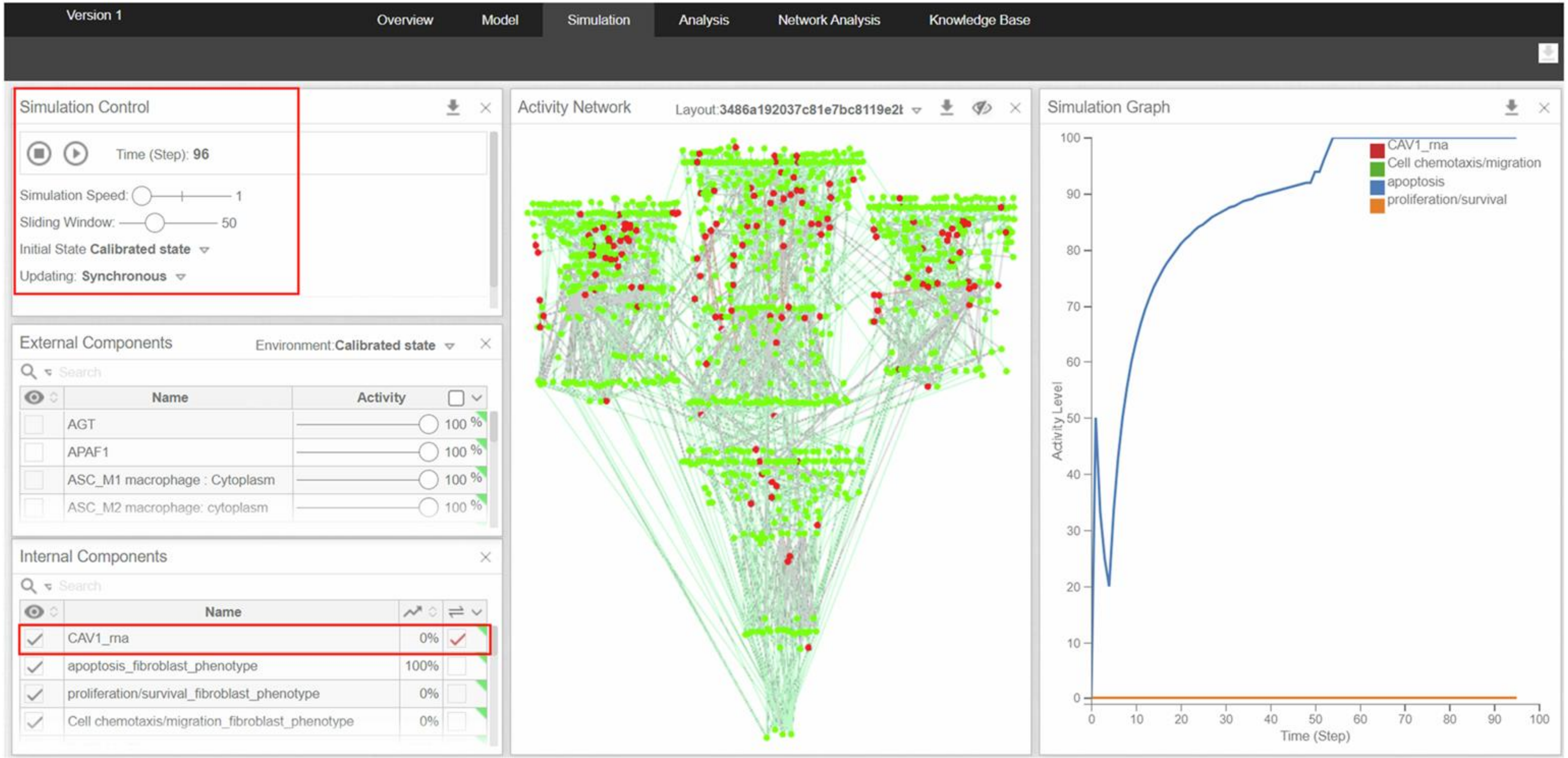


(**Zerrouk** et al., npj Digital Medicine, 2024)

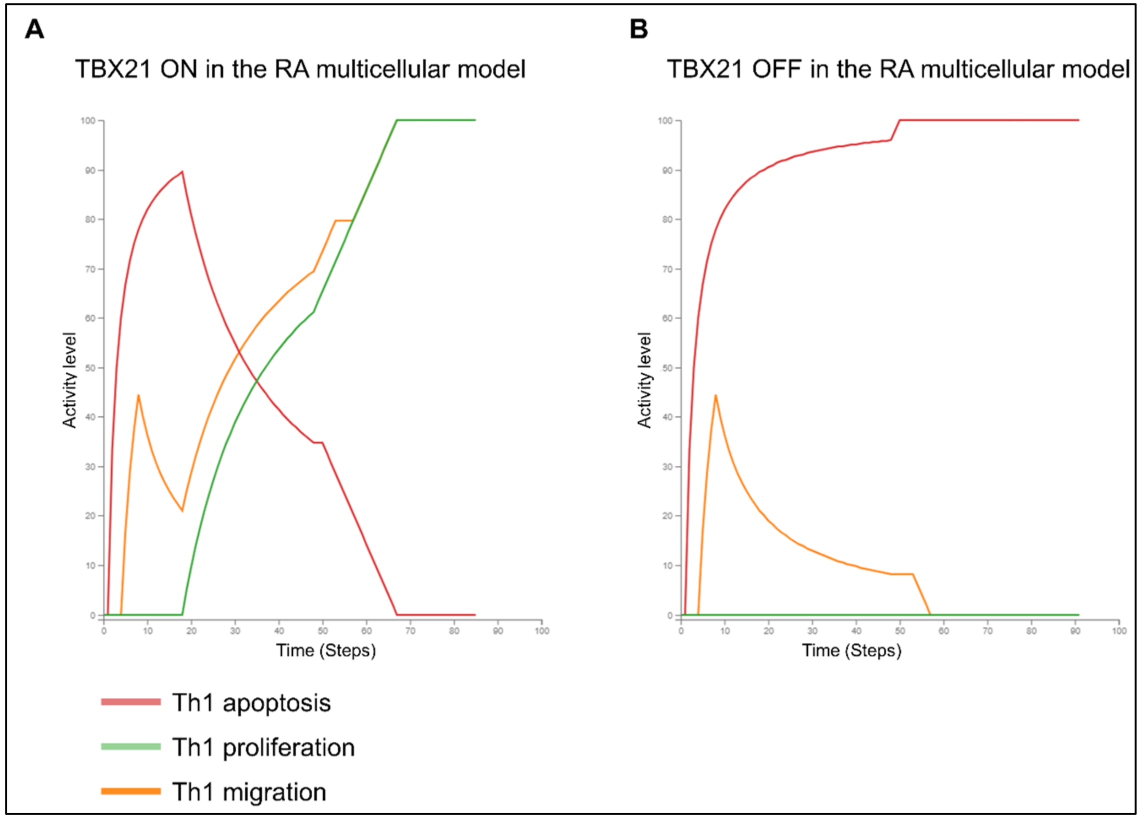
Size of published Boolean models since 2020

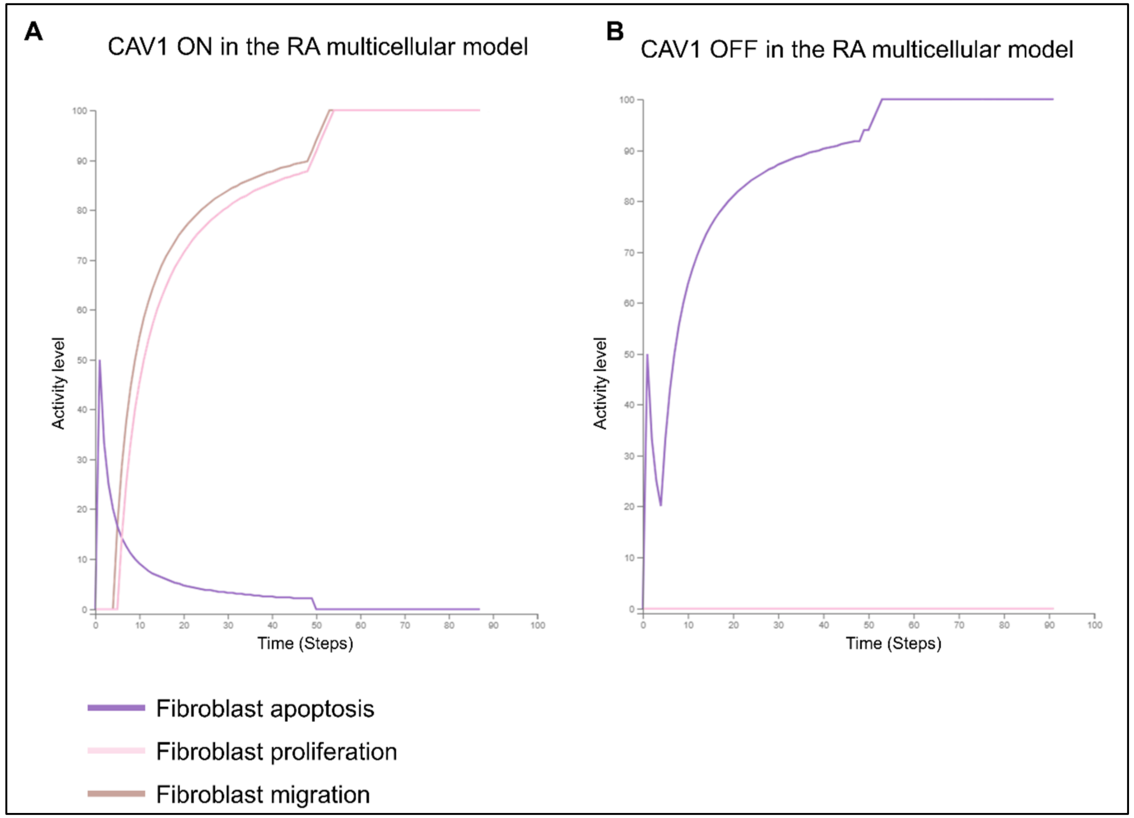
Publication	Number of nodes
Zerrouk et al, 2024	254
Zerrouk et al, 2024	309
Aghakhani et al., 2023	463
Aghakhani et al., 2022	359
Singh et al., 2022	321
Dnyane et al., 2021	265
Howell et al., 2022	175
Montagud et al., 2022	133
Corral-jara et al, 2021	82
Park et al., 2021	77
Bolouri et al., 2020	64
Cacace et al., 2020	56
Howell et al., 2023	38
de León et al., 2021	29
Ordaz-Arias et al., 2022	29
Gupta et al., 2023	26
Chapman et al.,2023	26

In silico simulations



In silico simulations using the RA multicellular model

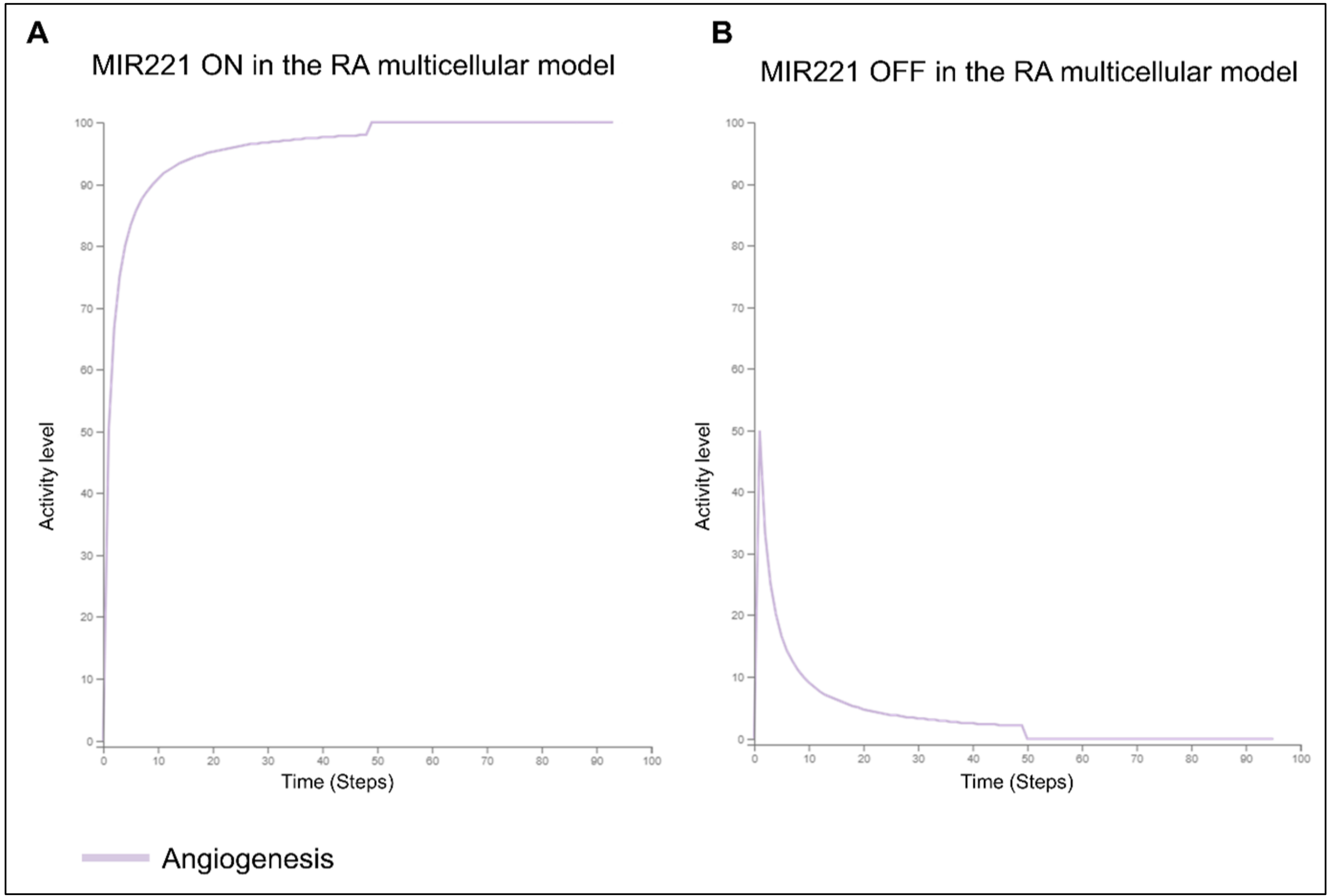




Published experimental evidence support **TBX21** inhibition for **downregulating hyperactive RA Th1** (Xue et al., 2014, Bruyn et al., 2007)

Published literature support **CAV1** inhibition for **downregulating hyperactive RA fibroblasts** (Xing et al., 2016, Li et al., 2017, Takeba et al., 2000)

In silico simulations using the RA multicellular model

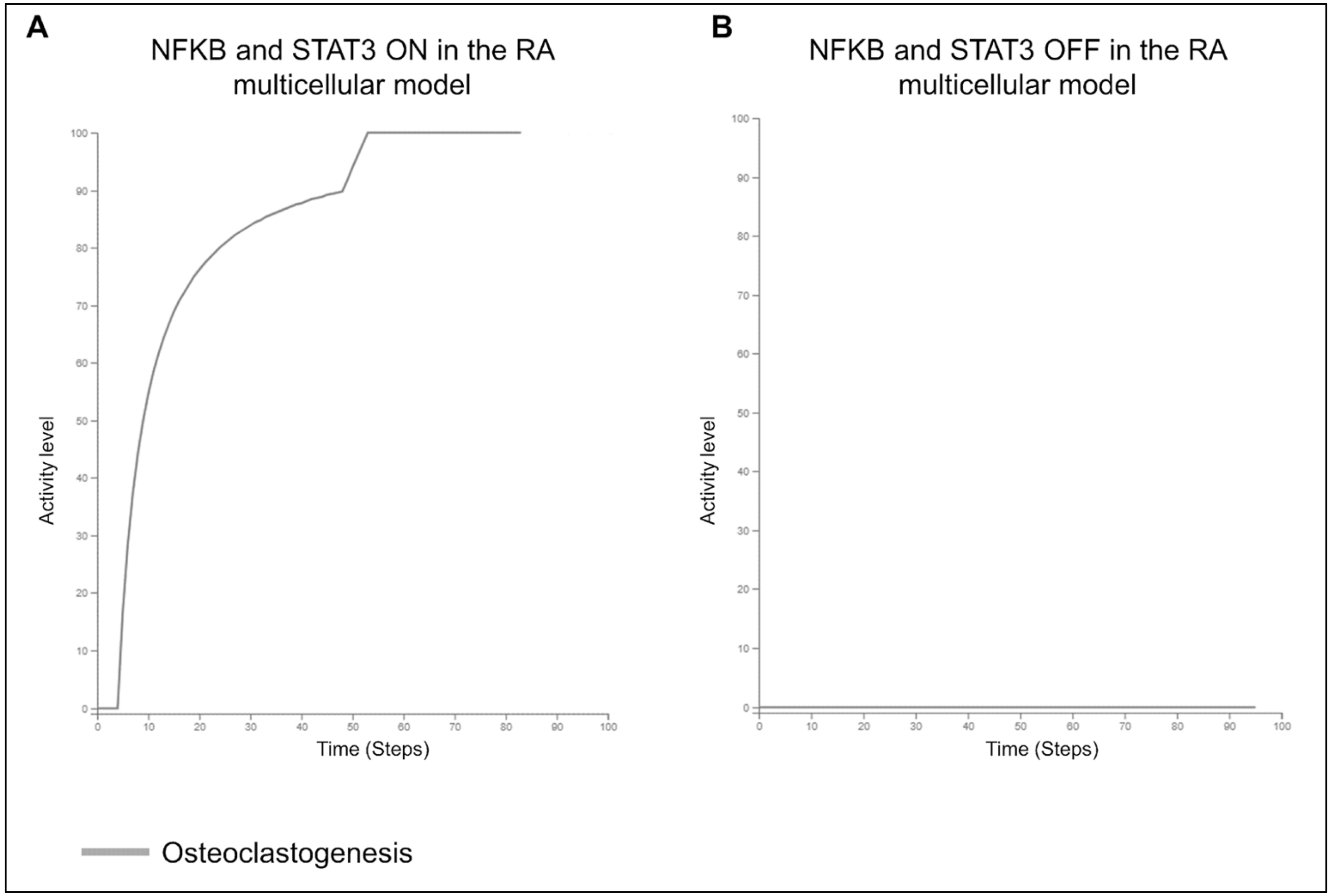


Experiments show that **MIR221** inhibition could reduce the **angiogenesis** in the arthritic joint (Mcmorrow et al., 2013)

⬇

MIR221 inhibition ➡ Subsequent activation of THBS1

New potentially therapeutic combinations



NF-κB and STAT3 double KO inhibited the differentiation of osteoclast precursor cells in the RA multicellular model

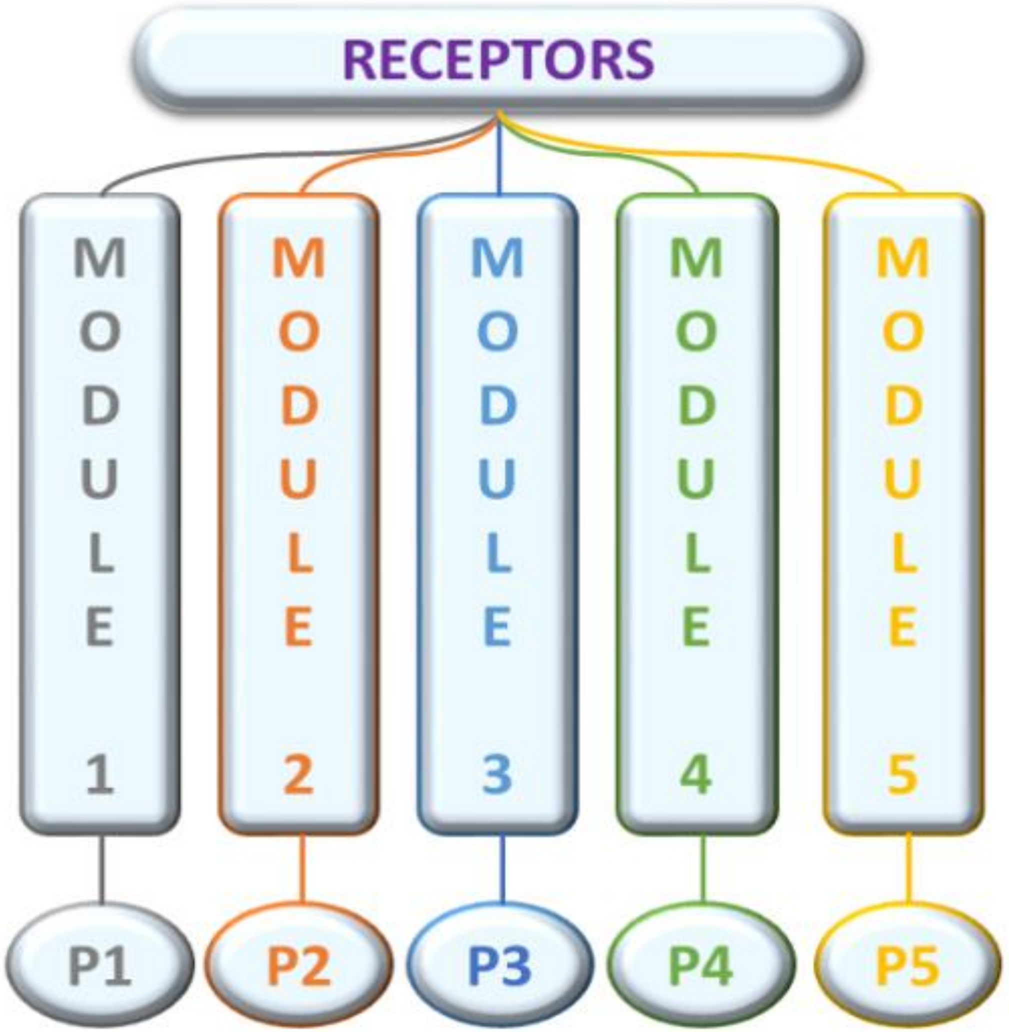
NF-κB and STAT3 is not associated with literature supporting its potential value as therapeutic target

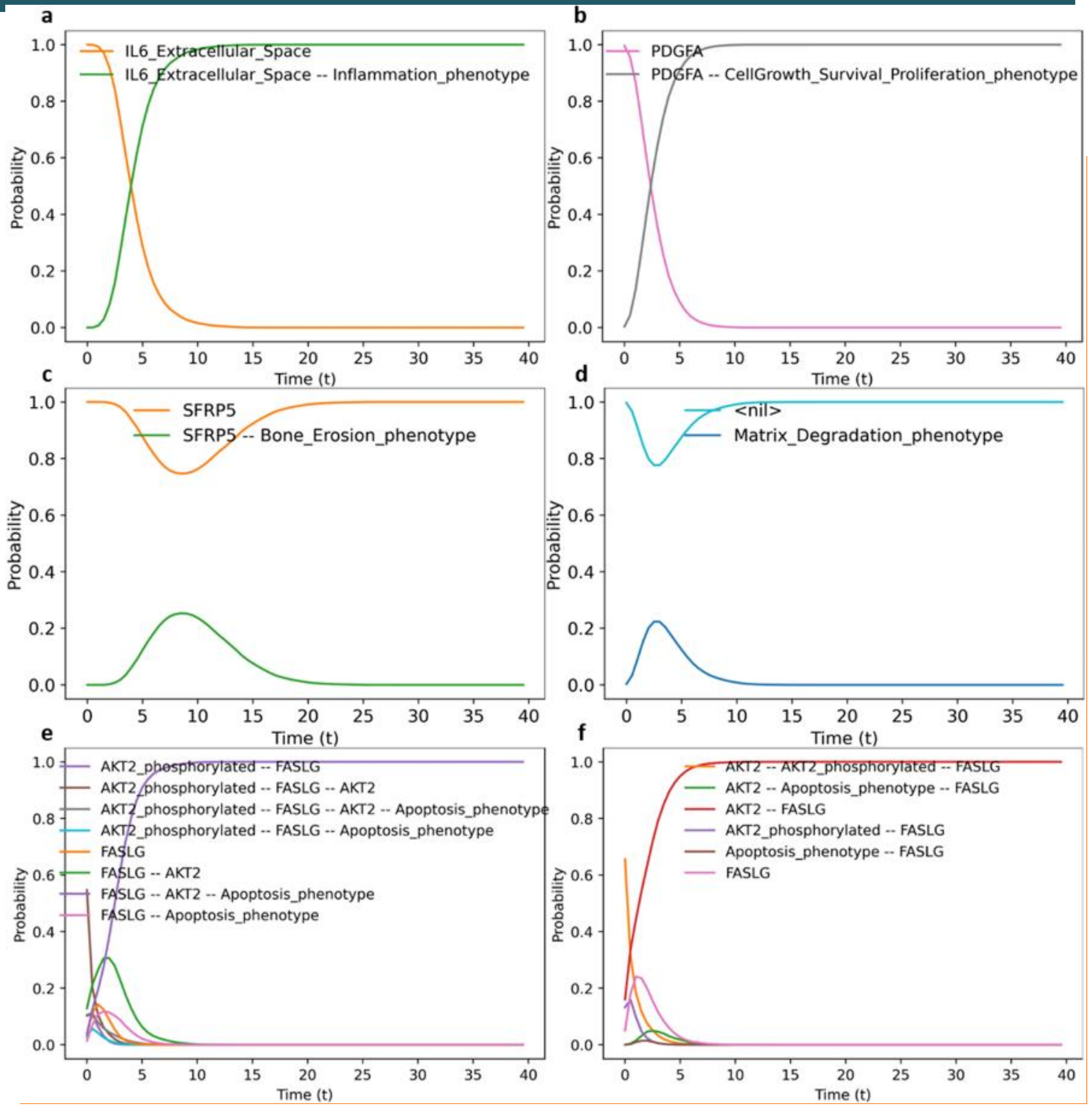
Targets and repurposed drugs

Successful targets	RA multi-cellular model's phenotypes														
	M1 Macrophage			M2 macrophage		Fibroblast			Th1			Joint			
	Apoptosis	Proliferation	Osteoclastogenesis	Apoptosis	Proliferation	Apoptosis	Proliferation	Migration	Apoptosis	Proliferation	Migration	Inflammation	Angiogenesis	Matrix degradation	Osteoclastogenesis
AKT2							↘	↘							
CAV1						↗	↘	↘							
CREB1							↘								
GSK3β				↘	↗										
ERK1		↘	↘												
MIR221													↘		
mTOR									↗	↘	↘				
NF-κB	↗	↘	↘					↘		↘	↘			↘	
TBX21									↗	↘	↘				

Successful targets	Target type	Associated disease(s)	Drugs with the highest status
CAV1	Literature-reported target	/	/
AKT2	Literature-reported target	/	Akt inhibitor VIII (Investigative)
CREB1	Literature-reported target	/	/
NF-kappaB	Successful target	Irritable bowel syndrome, Rheumatoid arthritis, Choreiform disorder, Lupus erythematosus, Multiple sclerosis...	Sulfasalazine (Approved)
TBX21	Literature-reported target	/	/
MTOR	Successful target	Arteries/arterioles disorder, Chronic myelomonocytic leukaemia, Hydrocephalus, Multiple myeloma, Renal cell carcinoma	Everolimus (Approved)
ERK1	Clinical trial target	Melanoma, Pancreatic cancer, Cancer, Arteries/arterioles disorder, Mature T-cell lymphoma	BVD-523 (Phase 2)
GSK3β	Clinical trial target	Myotonic disorder, Acute myeloid leukaemia, Osteosarcoma, Fragile X chromosome, Myeloproliferative neoplasm	Tideglusib (Phase 2/3)
MIR221	Literature-reported target	/	/

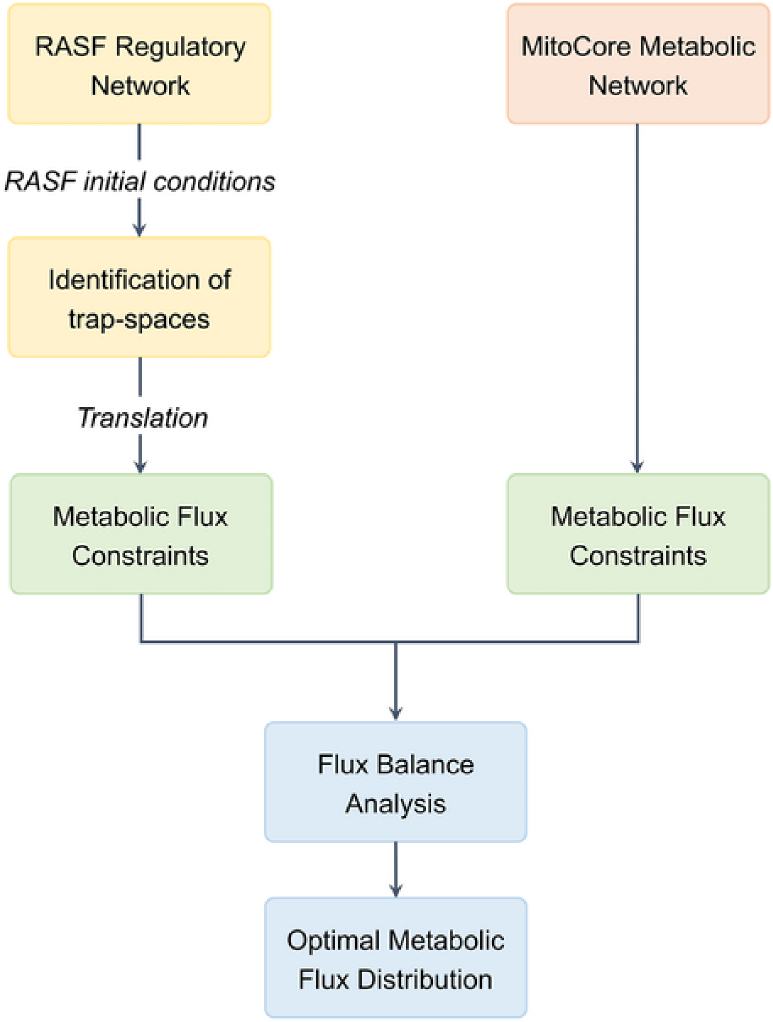
A large-scale Boolean model for RA-FLS

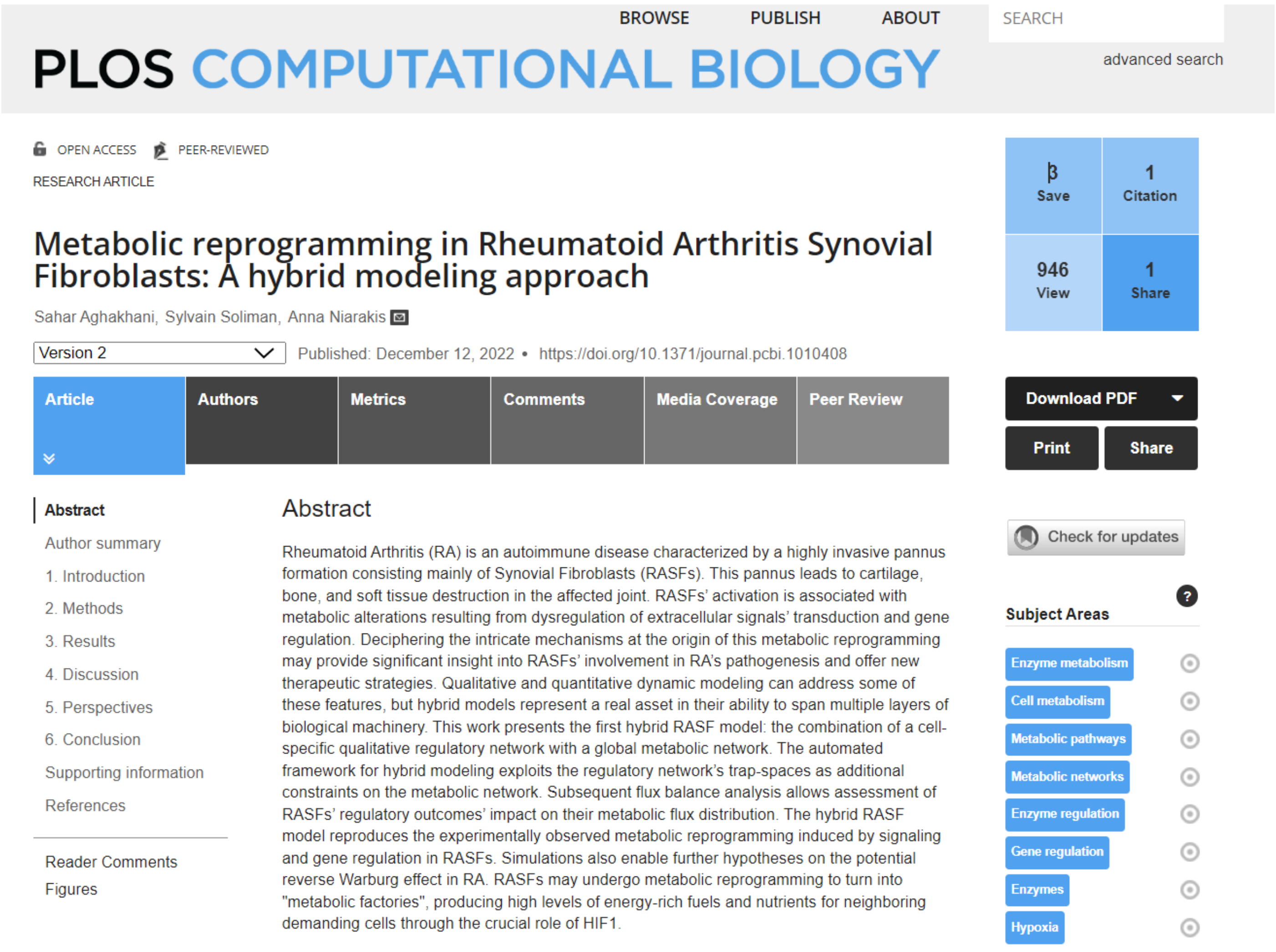




Vidisha Singh, Aurelien Naldi, Sylvain Soliman, Anna Niarakis, A modular, large-scale Boolean model of the Rheumatoid Arthritis fibroblast-like synoviocytes, *npj Systems Biology & Applications, 2023*

A hybrid model for RASF

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RESEARCH ARTICLE

Metabolic reprogramming in Rheumatoid Arthritis Synovial Fibroblasts: A hybrid modeling approach

Sahar Aghakhani, Sylvain Soliman, Anna Niarakis

Version 2 Published: December 12, 2022 • https://doi.org/10.1371/journal.pcbi.1010408

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Abstract
Author summary
1. Introduction
2. Methods
3. Results
4. Discussion
5. Perspectives
6. Conclusion
Supporting information
References

Reader Comments
Figures

Abstract

Rheumatoid Arthritis (RA) is an autoimmune disease characterized by a highly invasive pannus formation consisting mainly of Synovial Fibroblasts (RASFs). This pannus leads to cartilage, bone, and soft tissue destruction in the affected joint. RASFs' activation is associated with metabolic alterations resulting from dysregulation of extracellular signals' transduction and gene regulation. Deciphering the intricate mechanisms at the origin of this metabolic reprogramming may provide significant insight into RASFs' involvement in RA's pathogenesis and offer new therapeutic strategies. Qualitative and quantitative dynamic modeling can address some of these features, but hybrid models represent a real asset in their ability to span multiple layers of biological machinery. This work presents the first hybrid RASF model: the combination of a cell-specific qualitative regulatory network with a global metabolic network. The automated framework for hybrid modeling exploits the regulatory network's trap-spaces as additional constraints on the metabolic network. Subsequent flux balance analysis allows assessment of RASFs' regulatory outcomes' impact on their metabolic flux distribution. The hybrid RASF model reproduces the experimentally observed metabolic reprogramming induced by signaling and gene regulation in RASFs. Simulations also enable further hypotheses on the potential reverse Warburg effect in RA. RASFs may undergo metabolic reprogramming to turn into "metabolic factories", producing high levels of energy-rich fuels and nutrients for neighboring demanding cells through the crucial role of HIF1.

Check for updates

Subject Areas

Enzyme metabolism
Cell metabolism
Metabolic pathways
Metabolic networks
Enzyme regulation
Gene regulation
Enzymes
Hypoxia

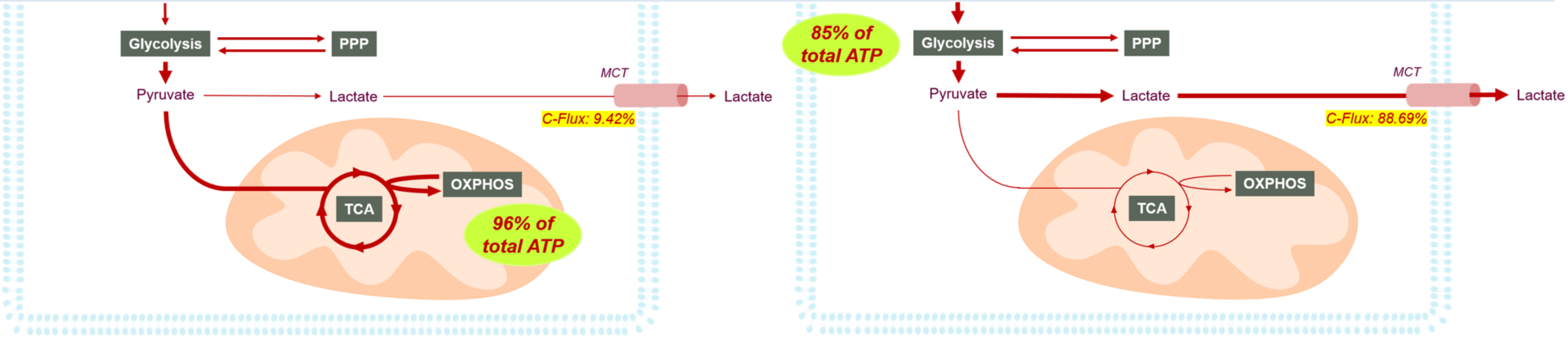
https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1010408

A hybrid model for RASF

Constraining FBA with Stable States Constraints

- **8 trap-spaces** identified;
- **12 metabolite's stable states = 0** (out of 29 common metabolites between RASF-Model and MitoCore);
- **7 metabolic enzyme's stable states = 0** (out of 19 common enzymes between RASF-Model and MitoCore);

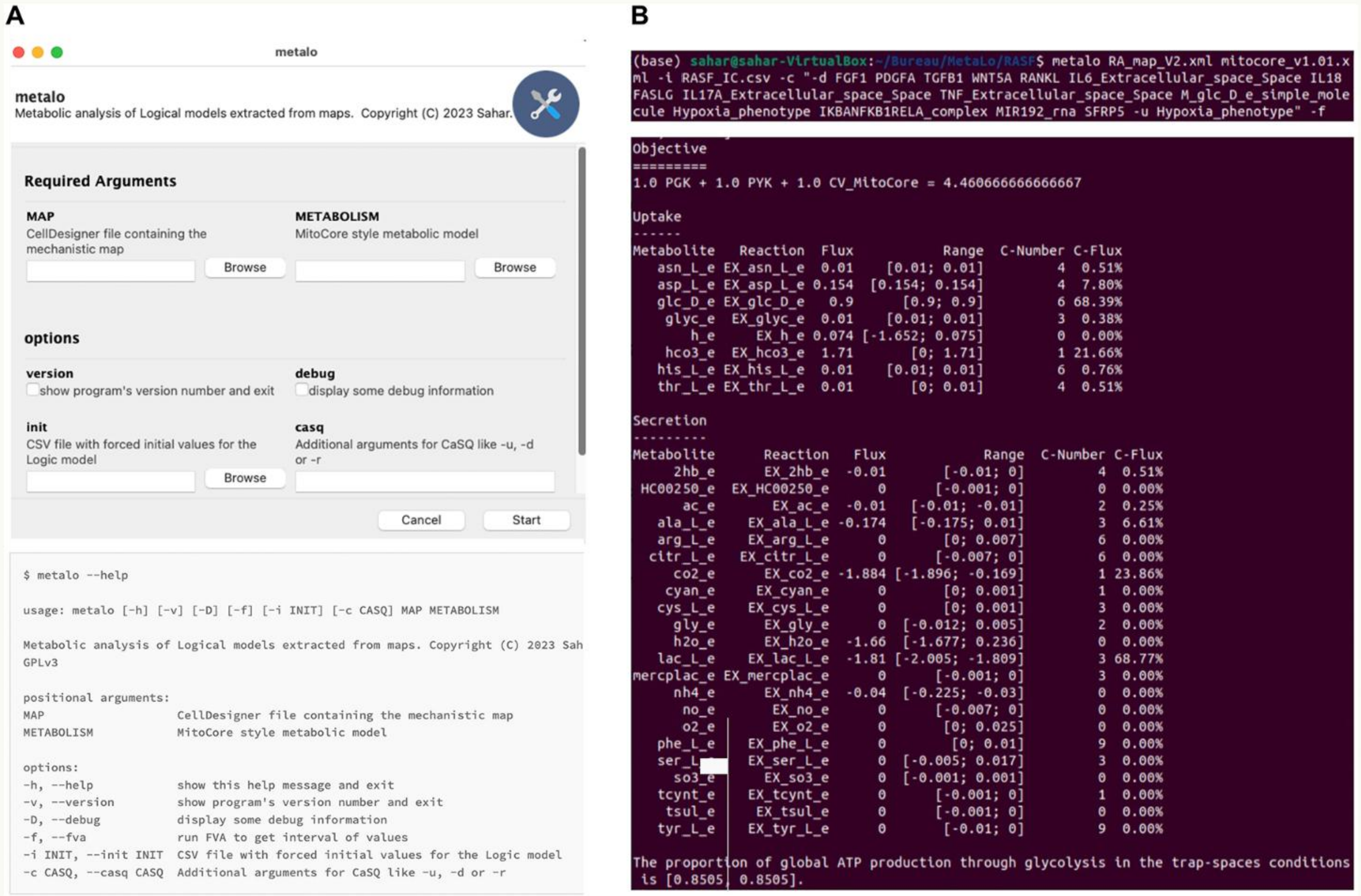
⇒ **50 metabolic reactions' flux constrained to 0** from the regulatory network.



MetaLo: metabolic analysis of Logical models extracted from molecular interaction maps

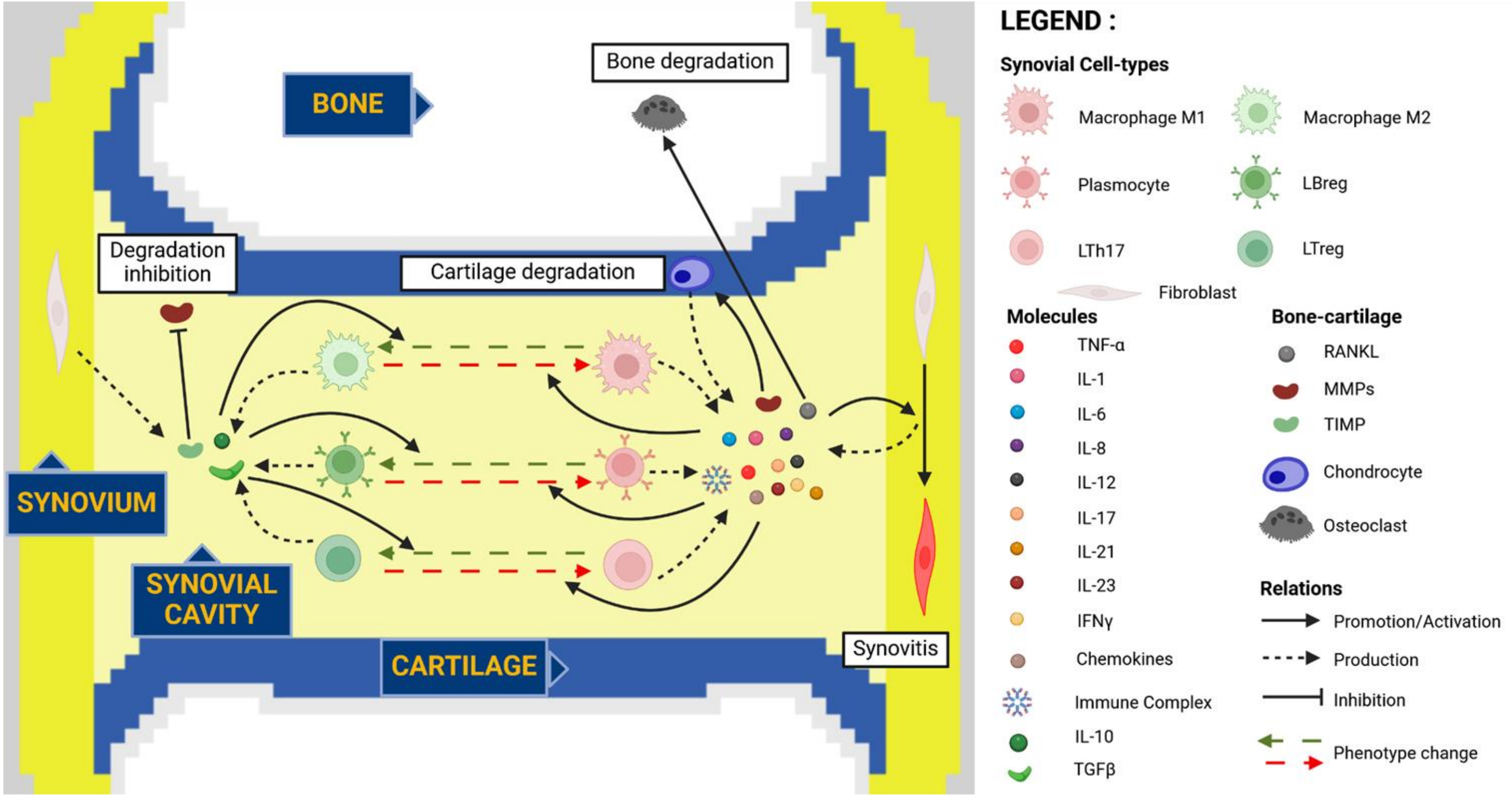
Sahar Aghakhani, Anna Niarakis and Sylvain Soliman

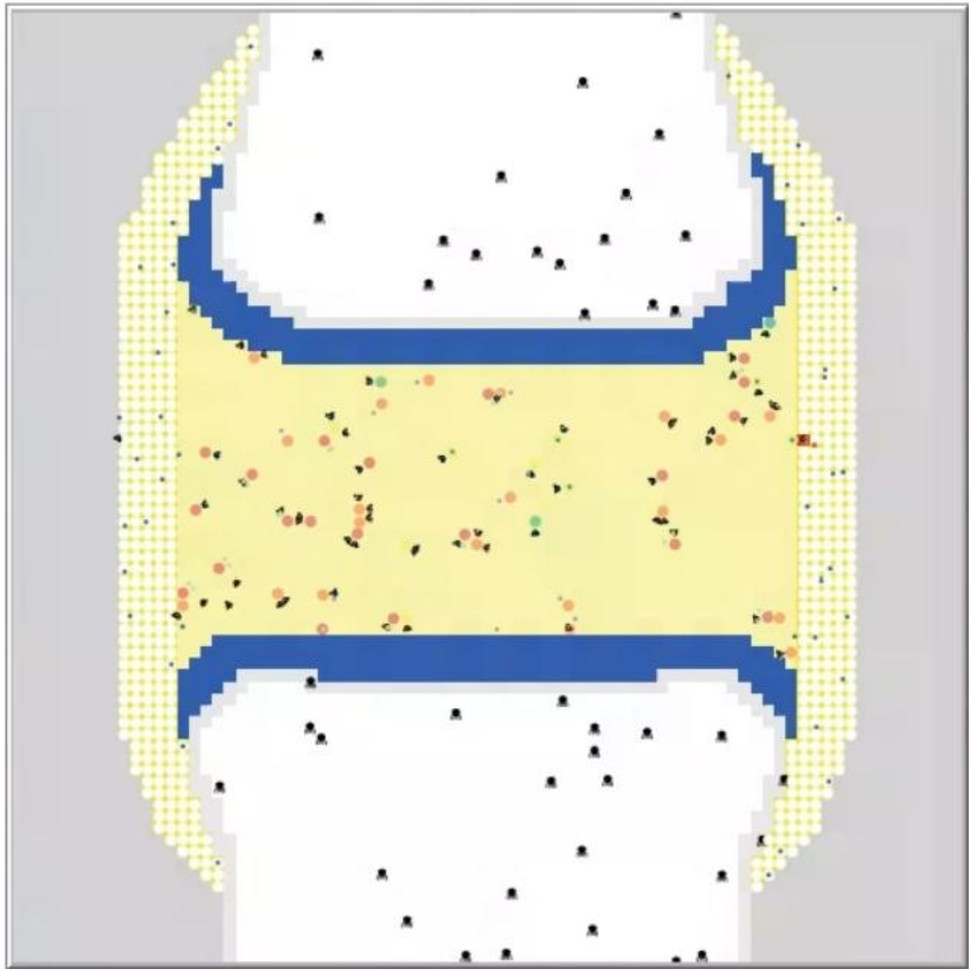
Published/Copyright: February 6, 2024



https://pypi.org/project/metalo/

Agent-Based Modelling





Main actors

Pro-inflammatory

Contributes to the inflammatory cycle and activates reactive components

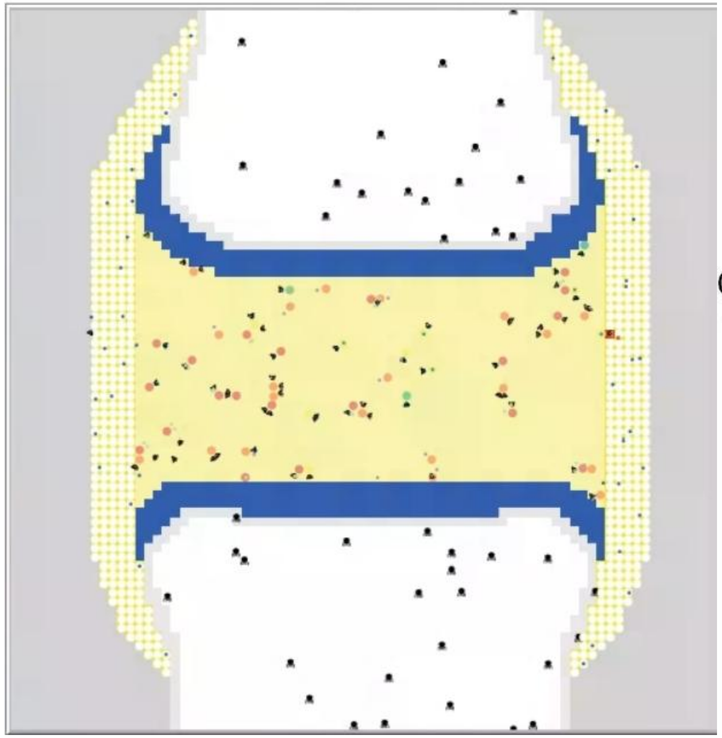
Reactive

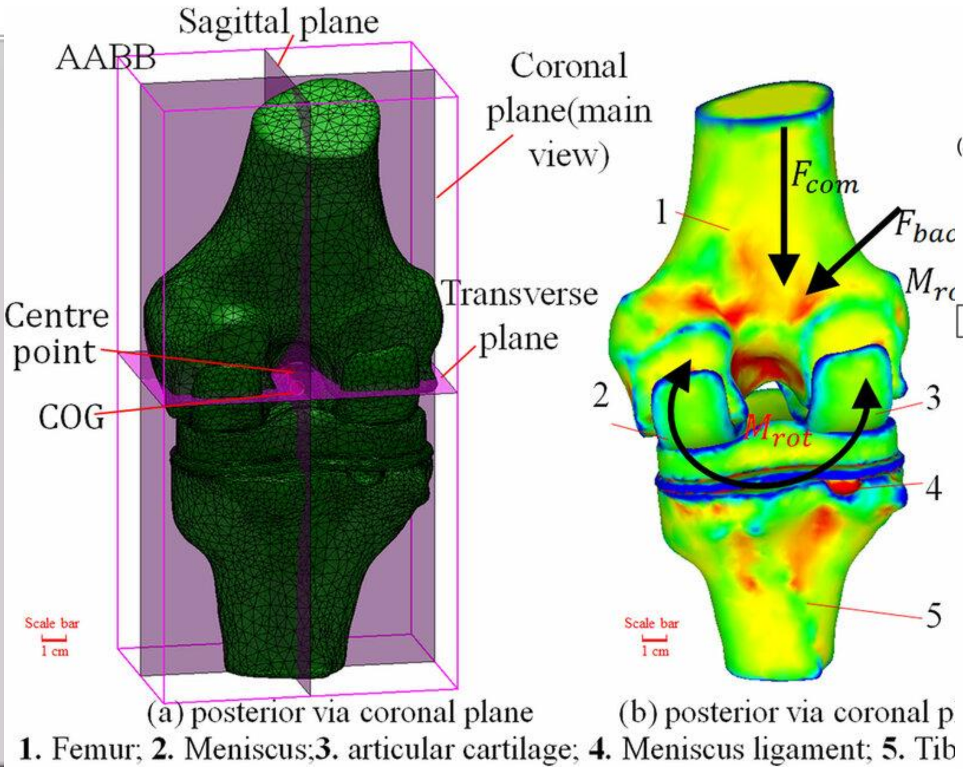
Depends on the signals of other components, can better or worsen the inflammation and degradation

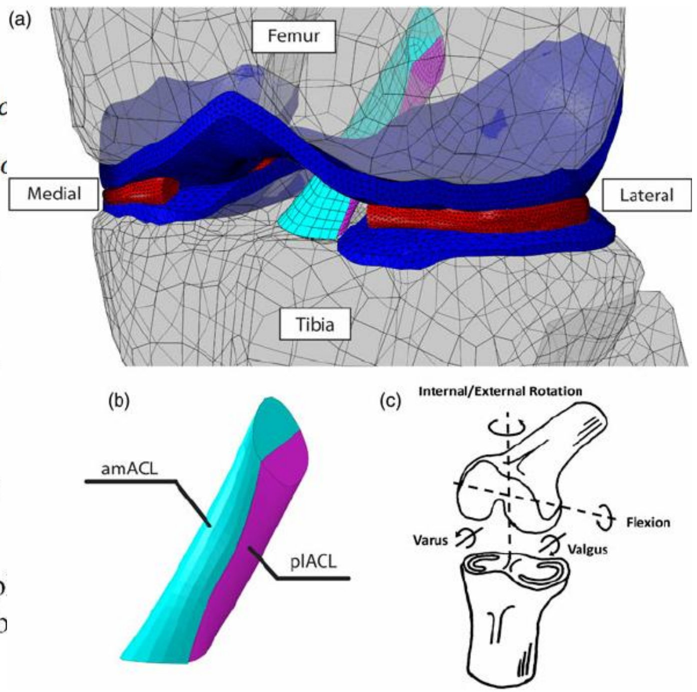
Anti-inflammatory

Helps alleviating the immune response and tends to help keeping the system's integrity

Reconcile micro & macro scales







ABM of RA knee joint (scRNAseq-guided prototype)

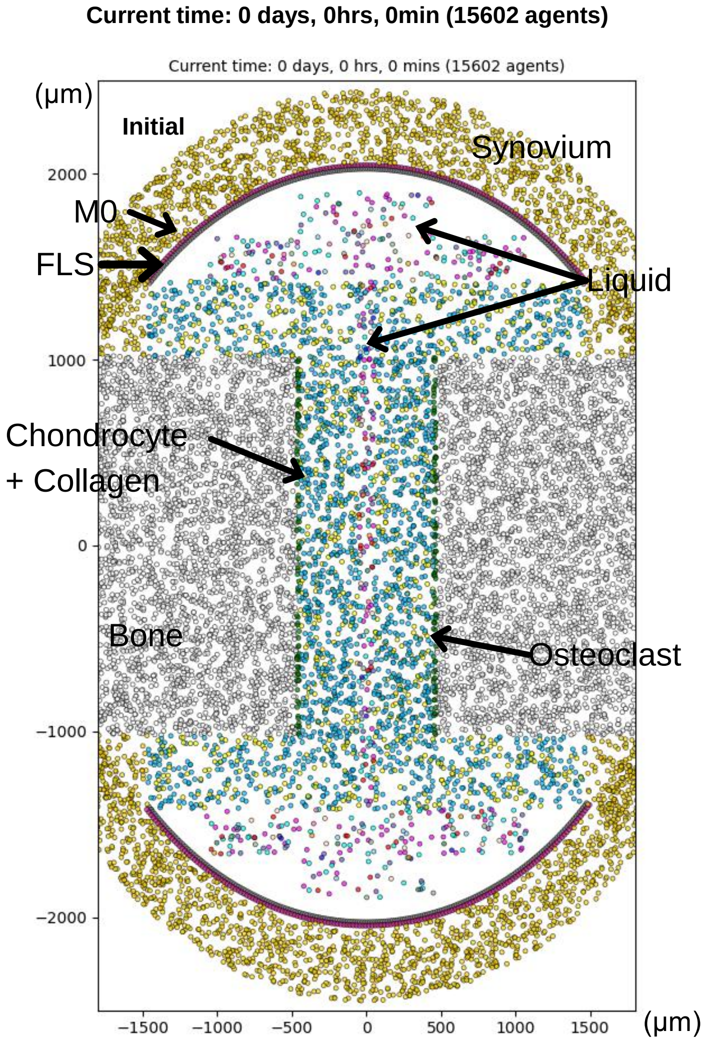


Figure 1 : *Initial conditions for the agent-based model (ABM) of a rheumatoid arthritis knee joint, based on the conceptual model. To simplify the simulation, cell infiltration via blood vessels is not modeled. The system is initialized after infiltration, with a pre-swollen synovial fluid.*

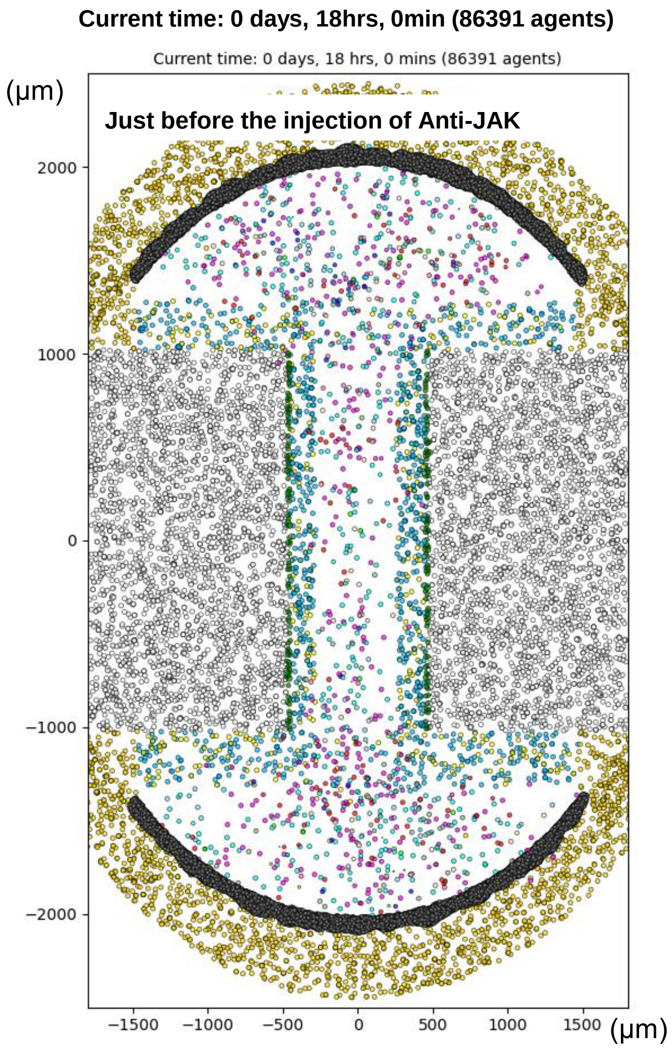


Figure 2 : *Evolution of the joint subject to RA, 18 hours after the beginning of the simulation. The cartilage die, the fibroblasts increase, cells of the liquid are multiplied. 5 mg are injected*

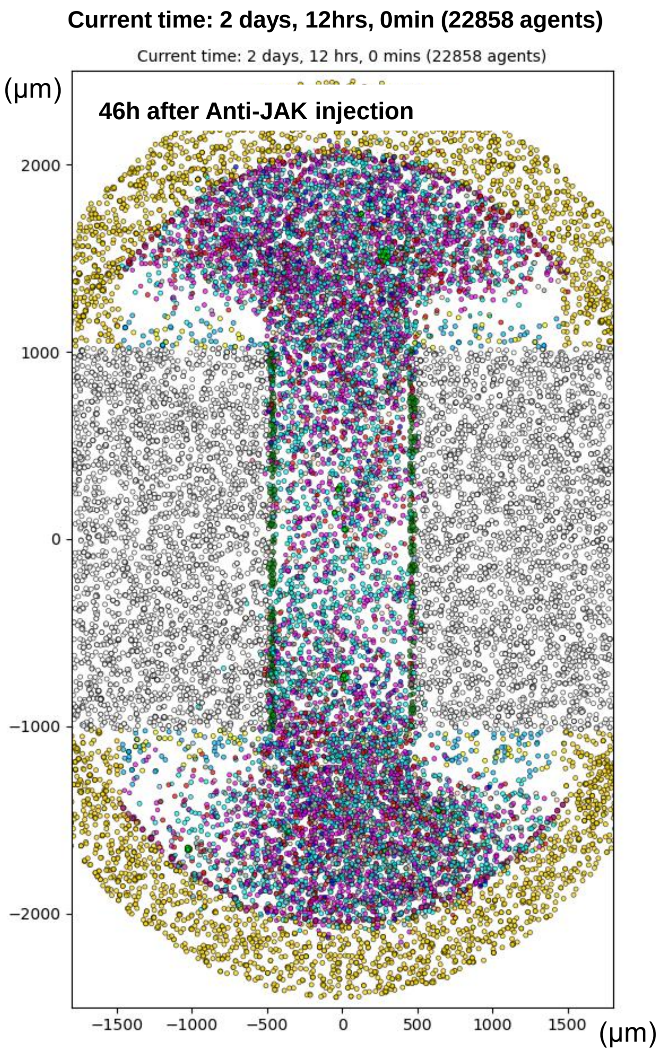


Figure 3 : *Evolution of the joint subject to RA, 46 hours after the injection of the Anti-JAK, Tofacitinib. The quantity of fibroblast decrease, however cartilage continue to die and cell liquid to multiply.*

CoSysBio; unpublished data

Results on the evolution of the model

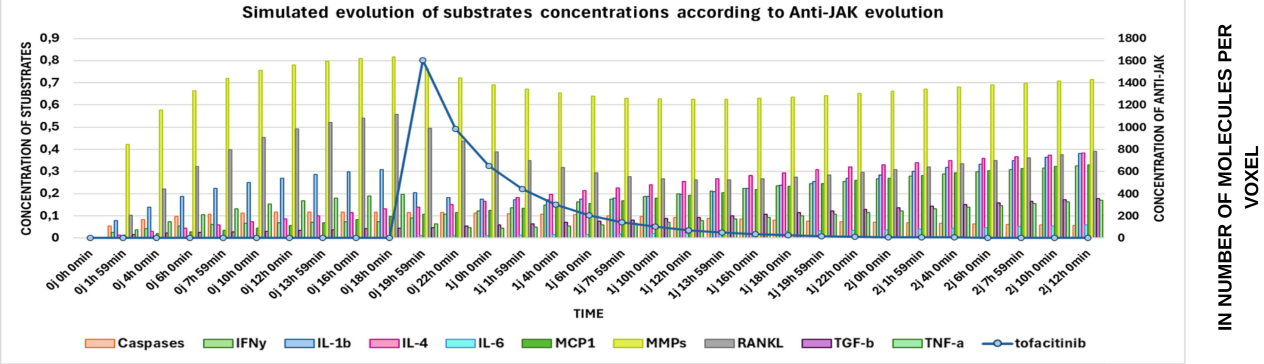


Figure 1 : *Evolution of the concentration of the substrates present in a rheumatoid arthritis joint according to the evolution of Tofacitinib concentration. Once the model is calibrated this result will allow to adjust the quantity and rate of injection of Tofacitinib according to the patient need to reduce inflammatory substrates.*

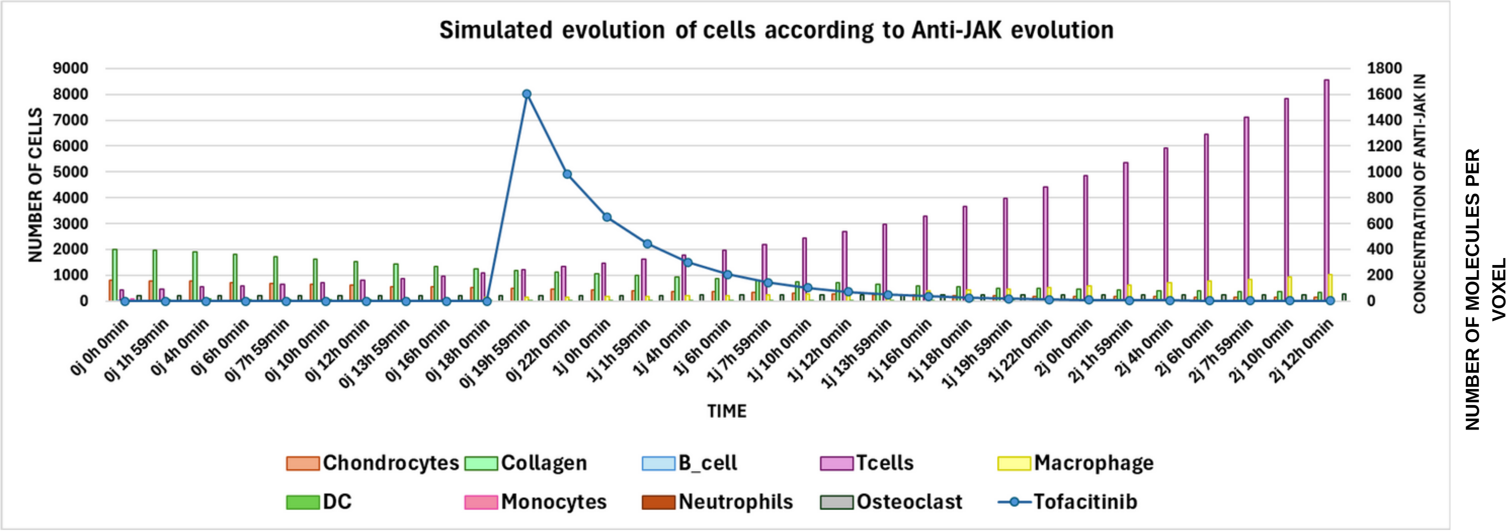
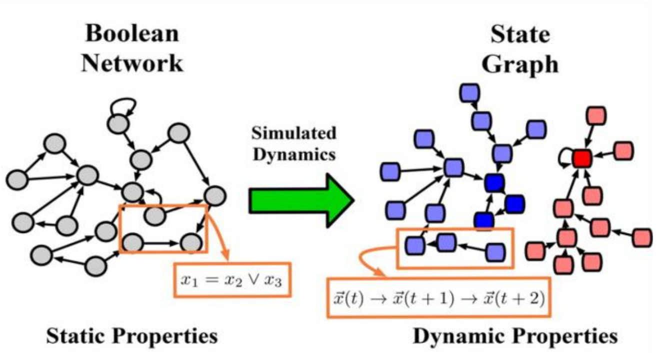


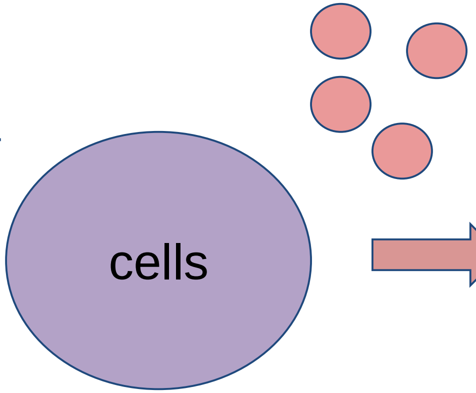
Figure 2 : *Evolution of the number of cells present in a rheumatoid arthritis joint according to the evolution of Tofacitinib concentration.*

Nested ABMs – The Babushka effect

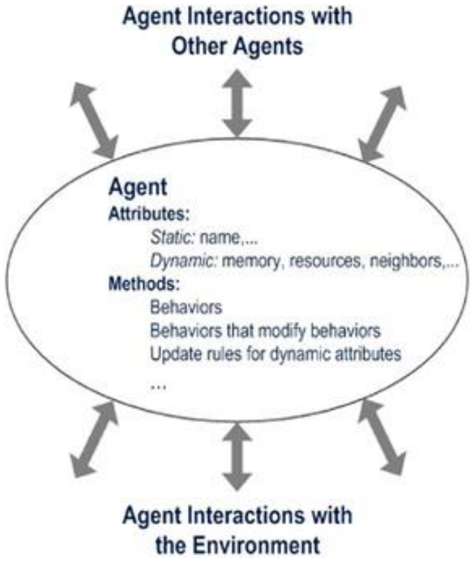
Intracellular networks
modelled as BNs



Decision-making systems for the agents



Cells and secreted molecules modelled as agents



Rheumatoid Arthritis

npj | systems biology and applications

Article | Open access | Published: 15 July 2023

A large-scale Boolean model of the rheumatoid arthritis fibroblast-like synoviocytes predicts drug synergies in the arthritic joint

Vidisha Singh, Aurelien Naldi, Sylvain Soliman & Anna Niarakis

npj Systems Biology and Applications 9, Article number: 33 (2023) | Cite this article

Abstract

Rheumatoid arthritis (RA) is a complex autoimmune disease with an unknown aetiology. However, rheumatoid arthritis fibroblast-like synoviocytes (RA-FLS) play a significant role in initiating and perpetuating destructive joint inflammation by expressing immuno-modulating cytokines, adhesion molecules, and matrix remodelling enzymes. In addition, RA-FLS are primary drivers of inflammation, displaying high proliferative rates and an apoptosis-resistant

npj | systems biology and applications

Article | Open access | Published: 26 January 2024

Large-scale computational modelling of the M1 and M2 synovial macrophages in rheumatoid arthritis

Naouel Zerrouk, Rachel Alcraft, Benjamin A. Hall, Franck Augé & Anna Niarakis

npj Systems Biology and Applications 10, Article number: 10 (2024) | Cite this article

Abstract

Macrophages play an essential role in rheumatoid arthritis. Depending on their phenotype (M1 or M2), they can play a role in the initiation or resolution of inflammation. The M1/M2 ratio in rheumatoid arthritis is higher than in healthy controls. Despite this, no treatment targeting specifically macrophages is currently used in clinics. Thus, devising strategies to selectively deplete proinflammatory macrophages and promote anti-inflammatory macrophages

PLOS COMPUTATIONAL BIOLOGY

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RESEARCH ARTICLE

Metabolic reprogramming in Rheumatoid Arthritis Synovial Fibroblasts: A hybrid modeling approach

Sahar Aghakhani, Sylvain Soliman, Anna Niarakis

Version 2 Published: December 12, 2022 • https://doi.org/10.1371/journal.pcbi.1010408

Abstract

Rheumatoid Arthritis (RA) is an autoimmune disease characterized by a highly invasive pannus formation consisting mainly of Synovial Fibroblasts (RASFs). This pannus leads to cartilage, bone, and soft tissue destruction in the affected joint. RASFs' activation is associated with metabolic alterations resulting from dysregulation of extracellular signals' transduction and gene regulation. Deciphering the intricate mechanisms at the origin of this metabolic reprogramming may provide significant insight into RASFs' involvement in RA's pathogenesis and offer new therapeutic strategies. Qualitative and quantitative dynamic modeling can address some of these features, but hybrid models represent a real asset in their ability to span multiple layers of biological machinery. This work presents the first hybrid RASF model: the combination of a cell-specific qualitative regulatory network with a global metabolic network. The automated framework for hybrid modeling exploits the regulatory network's trap-spaces as additional constraints on the metabolic network. Subsequent flux balance analysis allows assessment of RASFs' regulatory outcomes' impact on their metabolic flux distribution. The hybrid RASF model reproduces the experimentally observed metabolic reprogramming induced by signaling and gene regulation in RASFs. Simulations also enable further hypotheses on the potential reverse Warburg effect in RA. RASFs may undergo metabolic reprogramming to turn into "metabolic factories", producing high levels of energy-rich fuels and nutrients for neighboring demanding cells through the crucial role of HIF1.

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MetaLo: metabolic analysis of Logical models extracted from molecular interaction maps

Sahar Aghakhani, Anna Niarakis and Sylvain Soliman

From the journal Journal of Integrative Bioinformatics
https://doi.org/10.1515/jib-2023-0048

Abstract

Molecular interaction maps (MIMs) are static graphical representations depicting complex biochemical networks that can be formalized using one of the Systems Biology Graphical Notation languages. Regardless of their extensive coverage of various biological processes, they are limited in terms of dynamic insights. However, MIMs can serve as templates for developing dynamic computational models. We present MetaLo, an open-source Python package that enables the coupling of Boolean models inferred from process description MIMs with generic core metabolic networks. MetaLo provides a framework to study the impact of signaling cascades, gene regulation processes, and metabolic flux distribution of central energy production pathways. MetaLo computes the Boolean model's asynchronous asymptotic behavior, through the identification of trap-spaces, and extracts metabolic constraints to contextualize the generic metabolic network. MetaLo is able to handle large-scale Boolean models and genome-scale metabolic models without requiring kinetic information or manual tuning. The framework behind MetaLo enables in depth analysis of the regulatory model, and may allow tackling a lack of omics data in poorly addressed biological fields to contextualize generic metabolic networks along with improper automatic reconstructions of cell- and/or disease-specific metabolic networks. MetaLo is available at https://pypi.org/project/metalo/ under the terms of the GNU General Public License v3.

ORIGINAL RESEARCH article

Front. Syst. Biol., 11 July 2022
Sec. Data and Model Integration
Volume 2 - 2022 | https://doi.org/10.3389/fsysb.2022.925791

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A Mechanistic Cellular Atlas of the Rheumatic Joint

Naouel Zerrouk[1,2†] Sahar Aghakhani[1,3†] Vidisha Singh[1] Franck Augé[2] Anna Niarakis[1,3*]

[1] GenHotel—Laboratoire Européen de Recherche Pour La Polyarthrite Rhumatoide, University Paris-Saclay, University Evry, Evry, France
[2] Sanofi R&D, AI and Deep Analytics—Omics Data Science, Chilly-Mazarin, France
[3] Lifeware Group, Inria Saclay, Palaiseau, France

Rheumatoid Arthritis (RA) is an autoimmune disease of unknown aetiology involving complex interactions between environmental and genetic factors. Its pathogenesis is suspected to arise from intricate interplays between signalling, gene regulation and metabolism, leading to synovial inflammation, bone erosion and cartilage destruction in the patients' joints. In addition, the resident synoviocytes of macrophage and fibrobl types can interact with innate and adaptive immune cells and contribute to the disease's debilitating sympto Therefore, a detailed, mechanistic mapping of the molecular pathways and cellular crosstalks is essential to understand the complex biological processes and different disease manifestations. In this regard, we preser the RA-Atlas, an SBGN-standardized, interactive, manually curated representation of existing knowledge rel to the onset and progression of RA. This state-of-the-art RA-Atlas includes an updated version of the globa

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Identification of putative master regulators in rheumatoid arthritis synovial fibroblasts using gene expression data and network inference

Naouel Zerrouk, Quentin Miagoux, Aurelien Dispot, Mohamed Elati & Anna Niarakis

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Abstract

Rheumatoid Arthritis

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Building a modular and multi-cellular virtual twin of the synovial joint in Rheumatoid Arthritis

Naouel Zerrouk, Franck Augé & Anna Niarakis

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Abstract

Rheumatoid arthritis is a complex disease marked by joint pain, stiffness, swelling, and chroni synovitis, arising from the dysregulated interaction between synoviocytes and immune cells. I unclear etiology makes finding a cure challenging. The concept of digital twins, used in engineering, can be applied to healthcare to improve diagnosis and treatment for complex diseases like rheumatoid arthritis. In this work, we pave the path towards a digital twin of the arthritic joint by building a large, modular biochemical reaction map of intra- and intercellula interactions. This network, featuring over 1000 biomolecules, is then converted to one of the largest executable Boolean models for biological systems to date. Validated through existing knowledge and gene expression data, our model is used to explore current treatments and identify new therapeutic targets for rheumatoid arthritis.

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Anna Niarakis, Reinhard Laubenbacher, Gary An, Yaron Ilan, Jasmin Fisher, Åsmund Flobak, Kristin Reiche, María Rodríguez Martínez, Liesbet Geris, Luiz Ladeira, Lorenzo Veschini, Michael L. Blinov, Francesco Messina, Luis L. Fonseca, Sandra Ferreira, Arnau Montagud, Vincent Noël, Malvina Marku, Eirini Tsirvouli, Marcella M. Torres, Leonard A. Harris, T. J. Sego, Chase Cockrell, Amanda E. Shick, ... James A. Glazier + Show authors

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Abstract

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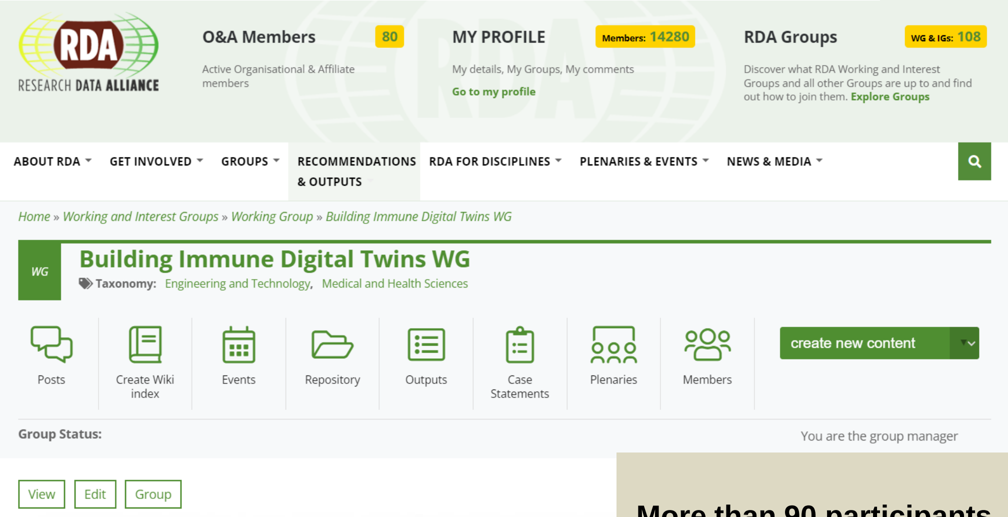
Abstract

Building Immune Digital Twins



More than 90 participants
19 countries: France, Netherlands, USA, India, Spain, UK, Norway, South Africa, Switzerland, Luxembourg, Greece, Denmark, Israel, Germany, Spain, Portugal, Belgium, Italy, Sweden

Interdisciplinary backgrounds:
Immunologists, clinicians, engineers, biologists, mathematicians, informaticians, bioinformaticians, modellers ,biochemists, computational biologists, pharmacists

Early career and senior scientists

Immune Digital Twins
A community effort to make IDTs a reality!

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About

Digital twins, customized simulation models pioneered in industry, are beginning to gain ground in medicine and healthcare, with some major successes in cardiovascular diagnostics and insulin pump control. Personalized computational models are also assisting in applications ranging from drug development to patient-tailored treatment optimization. Advanced medical digital twins will be essential to making personalized medicine a reality. Because the immune system plays an important role in such a wide range of diseases and health conditions, from fighting pathogens to autoimmune disorders, digital twins of the immune system (IDTs) will have an especially high impact. However, their development presents major challenges, stemming from the inherent complexity of the immune system and the difficulty of measuring many aspects of a patient's immune state in vivo. A collaborative interdisciplinary effort involving immunologists, clinicians, mathematical modelers, and software engineers is required to achieve substantial progress.

Important relevant readings

Regulatory circuits & Thomas' rules

- A **positive regulatory circuit** is **necessary** to generate **multiple stable states or attractors**
- A **negative regulatory circuit** is **necessary** to generate **sustained oscillatory behaviour**
- Thomas R (1988). *Springer Series in Synergics* **9**: 180-93.

Mathematical theorems and demonstrations:

- **In the differential framework:**
  - Thomas (+, 1994), Plathe *et al.* (±, 1995), Snoussi (±, 1998), Gouzé (±, 1998), Cinquin & Demongeot (+, 2002), Soulé (+, 2003).
- **In the discrete framework:**

  Aracena *et al.* (+, 2001), Remy *et al.* (±, 2005), Richard & Comet (+, 2005).